Replicating the St. Louis Fed DSGE Model

A Companion Note: Methodology, Estimation, Results, and Replication Assumptions

(US data, 1959Q1–2026Q1; Dynare implementation)

Replication note on Faria-e-Castro (2026)

September 3, 2026

Abstract

This note documents an independent replication of the St. Louis Fed DSGE model of Faria-e-Castro (2026). We (i) re-derive the model's optimality and equilibrium conditions from explicit microfoundations (Appendix A); (ii) implement the stationary system and a block-recursive steady state in Dynare; (iii) re-estimate the 63 free parameters by Bayesian MCMC on US data, 1959Q1–2026Q1; and (iv) reproduce the paper's headline objects: historical shock decompositions, forecast-error variance decompositions (FEVD), impulse responses, model-implied natural-rate (r^*) estimates against external measures, and unconditional and conditional forecasts, including the paper's Table 5. The replication is successful: 59 of 63 posterior means fall inside the paper's 10th-to-90th percentile bands, and the historical decompositions, FEVD, and impulse responses match. The remaining differences in the forecast table are each traced to one specific, defensible data choice, a data-anchored trend-growth estimate and a data-anchored inflation target, rather than to any error; output growth is additionally converted from the model's native per-capita object to the aggregate one the paper reports, and Table 3 shows that conversion in full. Section 5 collects the assumptions and pitfalls that a replicator should keep in mind; the Appendix reproduces the full derivations, from explicit microfoundations through detrending and the block-recursive steady state.

Contents

1 Introduction and scope

The St. Louis Fed DSGE model is a medium-scale New Keynesian model used as one input into forecasting and policy analysis at the Bank. Relative to the canonical Smets and Wouters (2007) framework it adds three ingredients: (i) two-agent (TANK) household heterogeneity between hand-to-mouth-leaning *workers* and *capitalists*; (ii) an explicit fiscal block with distortionary taxes, transfers, government debt, and behavioural tax/spending rules; and (iii) oil as both a production input and a consumption good, which lets the model separate core from headline inflation. The model is estimated by Bayesian methods on a long US sample and is used to produce historical narratives (shock decompositions), structural forecasts, and a model-consistent natural rate of interest. All model objects were produced in Dynare 7.1 with the estimation and post-estimation scripts described in Sections 2–4. Appendix A provides the full algebra behind the equilibrium conditions, keeping the paper's notation and equation numbers.

2 Methodology: the model

2.1 Structure

The economy is populated by a measure λ of workers and $1 - \lambda$ of capitalists. Both consume a CES bundle of core and oil goods (elasticity ν_C, oil weight ω_C) and hold government debt; capitalists additionally own the capital stock and all firms, choose investment and capital utilization, and receive firm profits, while workers face a portfolio adjustment cost $\Psi(B_t^w)$ that makes their consumption respond more strongly to current income (a smooth hand-to-mouth limit). Labor is intermediated by monopolistically competitive unions that set wages subject to Rotemberg adjustment costs, generating a wage Phillips curve; intermediate-goods producers combine a capital–labor composite with oil and set prices subject to Rotemberg costs, generating the price Phillips curve. A fiscal authority levies consumption, labor, and capital taxes, pays transfers, and issues debt, with labor- and capital-tax rates partially adjusting to the debt-to-GDP ratio; the central bank follows a smoothed Taylor rule. Government debt carries a convenience yield ϑ_t (as in Del Negro et al., 2017), which drives a wedge between the safe real rate and the return on capital. Two stochastic trends, labor-augmenting TFP Z_t and a labor-disutility shifter ξ_t, give the model its balanced-growth path. The full set of optimality conditions, their detrending onto the stationary system, and the steady state are derived in Appendix A.

2.2 The natural rate and the output gap

A distinctive output of the model is a model-consistent natural (flexible-price) allocation used to define the natural rate of interest r^* and the output gap. Following the paper's Section 2.12, we construct a parallel *flexible-price, flexible-wage* economy: the same real block with the Rotemberg price and wage frictions switched off, the price and wage markups held at their steady-state values $\bar{\mu}^p, \bar{\mu}^w$, and inflation held at the Fed's time-varying target, $\Pi_t^f = \Pi_t^*$. It shares the sticky economy's steady state, its shocks, and its predetermined states. The natural rate r_t^* is the safe real rate in this counterfactual economy, and the output gap is $100\,(\log GDP_t - \log GDP_t^f)$, the log distance between actual and flexible-price output. This block is *block-recursive*: it feeds nothing back into the sticky economy and therefore does not affect the likelihood, so it is computed as a post-estimation object (Section 4). Several convention choices in building it are discussed in Section 5.

2.3 The COVID episode

To prevent the extraordinary 2020–2021 observations from distorting the ordinary shock variances, the model introduces one-time COVID disturbances that *replace* three exogenous drivers during specified windows: the marginal-utility (demand) shifter χ_t, government transfers t_t, and the labor-disutility growth rate Γ_t^N (eqs. 112–114 of the paper). The χ and transfer shocks are one-time i.i.d.; the labor-disutility shock has an MA(1) structure. These are estimated by treating them as observables with missing values outside their active windows, so the Kalman smoother infers their magnitude without letting them propagate through the persistent AR components.

3 Estimation method

3.1 Data

The model is estimated on 15 quarterly US observables spanning 1959Q1–2026Q1: real per-capita GDP, consumption, and investment growth; government consumption and transfers growth; real wage growth; hours; the federal funds rate; core PCE inflation; a TFP series; the real oil and gasoline prices; long-run (10-year) inflation expectations; the 10-year Treasury yield; and a convenience-yield proxy (the spread between a safe long rate and the policy rate). Four series begin later than 1959Q1: hours (1964Q2), the 10-year Treasury yield and the convenience-yield proxy (both 1962Q1), and 10-year inflation expectations (1991Q4, the first quarter of the SPF's `CPI10` series). These, together with the COVID shocks, are handled with Dynare's missing-value capability in the Kalman filter. The expectations series is put on a PCE basis by subtracting a constant 0.5pp from the SPF's 10-year CPI expectation, following Del Negro et al. (2017).

3.2 Solution and priors

The model is solved by first-order perturbation around the deterministic steady state, coded with an explicit `steady_state_model` block implementing the block-recursive recipe of Appendix A, Section I. For speed the model is compiled to C (`use_dll`). Priors follow Smets and Wouters (2007) for the frictions and shock processes, with two model-specific choices carried over from the paper: tight priors around very high persistence ($\rho \approx 0.99$) for the inflation-target and convenience-yield AR coefficients (as in Del Negro et al., 2017, and Gelain and Lopez), to capture slow-moving movements in the Fed's implicit target and in the real rate; and priors placing the oil complementarities (ν_C, ν_Y) below one and close to zero (Gagliardone and Gertler, 2023).

3.3 Posterior sampling and diagnostics

The posterior is sampled by random-walk Metropolis–Hastings in three stages: (a) numerical mode-finding; (b) a short pilot run to tune the proposal and check mixing; and (c) a full run of eight parallel chains of 250,000 draws each, discarding the first 50% as burn-in, for one million posterior draws. The jumping covariance is the inverse Hessian at the mode and the proposal scale is the one the pilot tuned; the acceptance rate settled between 32.1% and 32.5% in every chain. Convergence is assessed with the Brooks–Gelman–Rubin multivariate and univariate diagnostics, Geweke

(1992) separated-means tests per chain, and per-block MCMC inefficiency factors; the marginal data density is computed by the modified harmonic mean. Post-estimation objects (the Kalman smoother, historical decompositions, FEVD, impulse responses, and conditional and unconditional forecasts) all reuse the stored MH draws (`load_mh_file`), so they reflect full posterior uncertainty without re-sampling. Because the natural-rate block is likelihood-neutral, it is appended only in the post-estimation stage.

4 Results

4.1 Parameter estimates

The replication reproduces the paper's posterior closely: 59 of the 63 estimated parameters have posterior means inside the paper's 10th-to-90th percentile band (its Tables 2 and 3), and the economically decisive frictions and policy coefficients essentially coincide. Table 1 lists a representative subset. Four parameters fall outside, and two of those only marginally: the price menu cost η_p (57.90 against a lower bound of 59.23) and the Taylor-rule output coefficient ϕ_Y (0.559 against 0.573) each miss by less than a twentieth of the band width. The two substantive misses are trend TFP growth Γ^Z (1.0039 against $[1.0043, 1.0055]$) and the persistence of the measurement error on ten-year inflation expectations ρ_{me,π^e} (0.712 against $[0.854, 0.995]$). Those are exactly the two channels behind the forecast-table differences discussed in Section 4.7: a data-anchored trend growth and a data-anchored inflation target.

One parameter deserves a note because it is the one the corrected utilization condition bears on directly. The utilization-cost curvature σ_a has a posterior mean of 0.496 with a 90% interval of $[0.342, 0.645]$, an implied posterior standard deviation of about 0.092 against a prior standard deviation of 0.10: the likelihood supplies roughly a sixth of the total precision, so σ_a is close to prior-determined. It is essentially unchanged by the tax-wedge correction to the utilization first-order condition described in Section 5, which moves κ_a mechanically by the factor $(1 - \bar{\tau}^k)$ but leaves the estimated dynamics where they were. That the data do not speak to this elasticity is itself the reason the correction is a matter of internal consistency rather than of fit.

Table 1: Selected posterior means: replication vs. paper (Faria-e-Castro, 2026, Table 2).

Parameter	Description	Replication	Paper
Γ^Z	TFP trend growth	1.0039	1.0049
$\bar{\vartheta}$	Convenience yield	1.0025	1.0021
$100(1/\beta - 1)$	Discount factor	0.0393	0.0383
ϕ_Π	Taylor rule, inflation	2.868	2.849
ϕ_Y	Taylor rule, output	0.559	0.787
ρ_r	Taylor rule, smoothing	0.810	0.806
σ_a	Utilization cost	0.496	0.488
α	Capital share	0.196	0.197
λ	Share of workers	0.543	0.588

4.2 Historical decompositions

Figures 1–4 reproduce the paper's simplified historical decompositions (its Figs. 4–5) over 2019Q4–2026Q1, grouping shocks into demand, supply, fiscal, monetary, and oil, with the COVID special shocks folded into their economic categories. As in the paper, the post-pandemic inflation surge is attributed largely to demand and fiscal forces with a direct monetary contribution (policy holding rates below the historically estimated Taylor prescription in 2022–2024), and the recovery closed the output gap and kept output above its flexible-price level from early 2021. The extended decompositions (Figs. 5–8, the paper's Figs. 9–10) break the same objects into the 12 individual structural shocks.

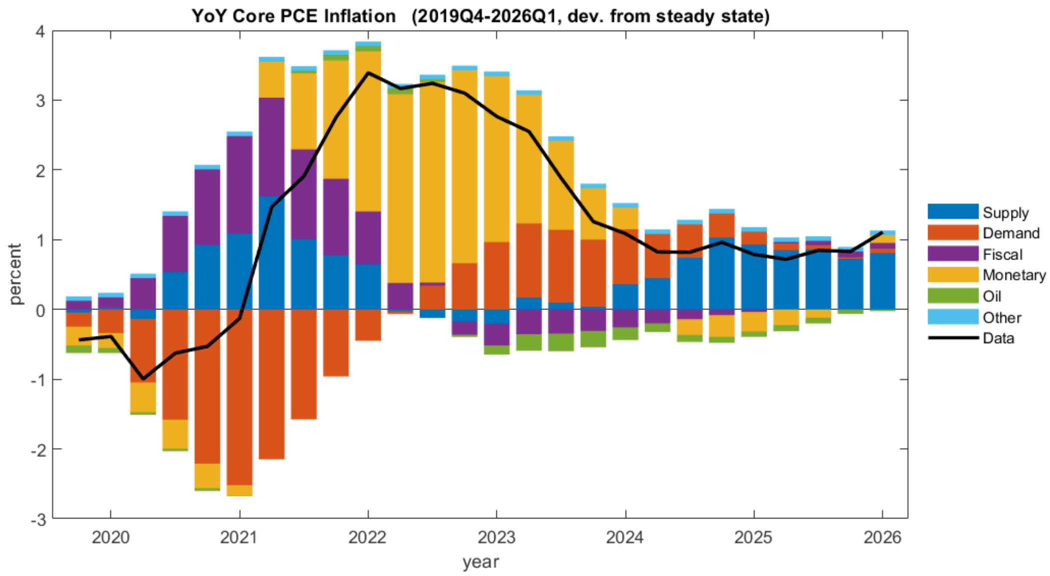

Figure 1: YoY core PCE inflation: historical decomposition, 5 shock groups (paper Fig. 4a).

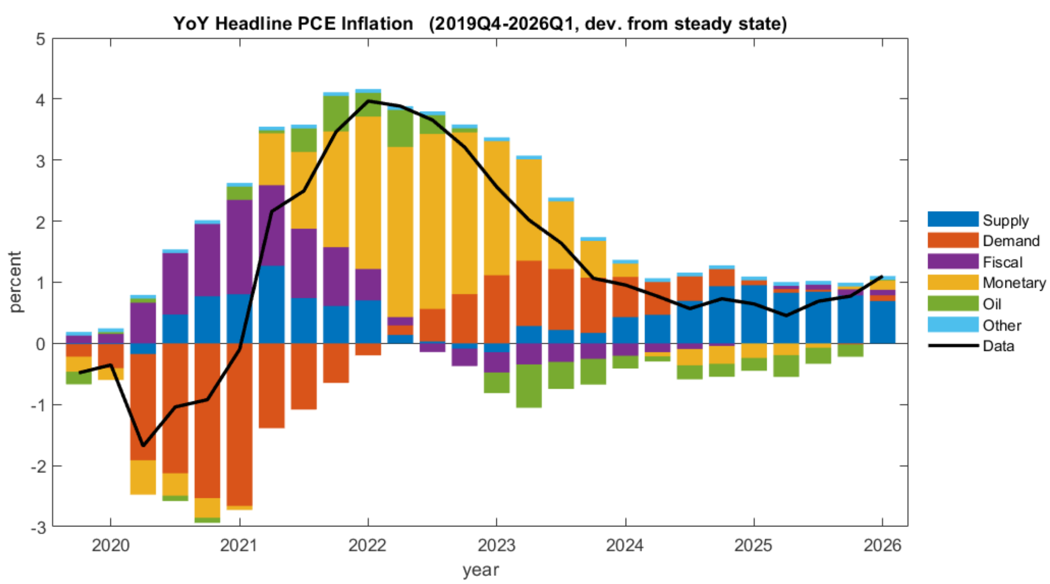

Figure 2: YoY headline PCE inflation: historical decomposition (paper Fig. 4b).

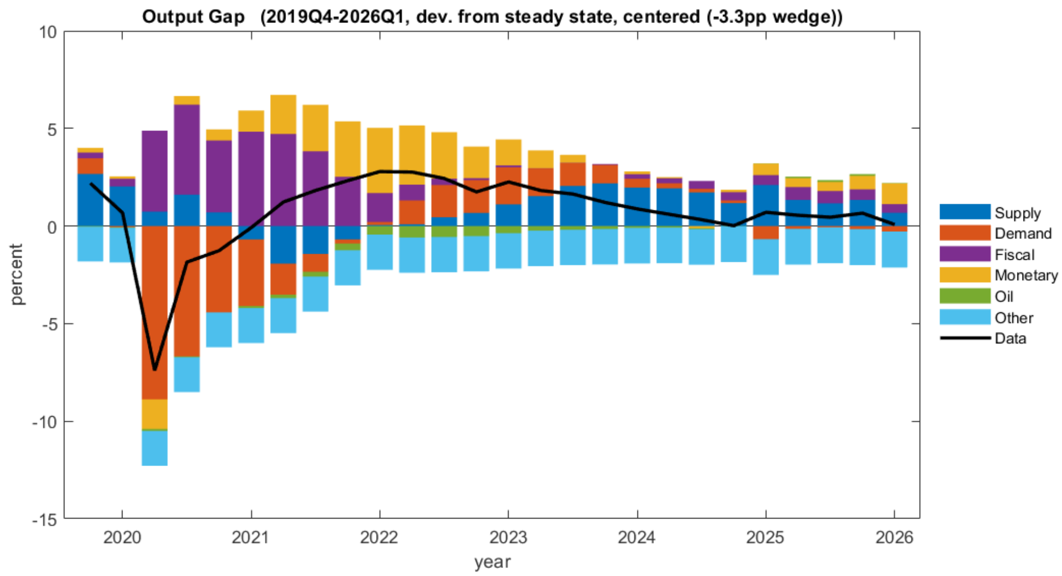

Figure 3: Output gap: historical decomposition, centered on the long-run level (paper Fig. 5a).

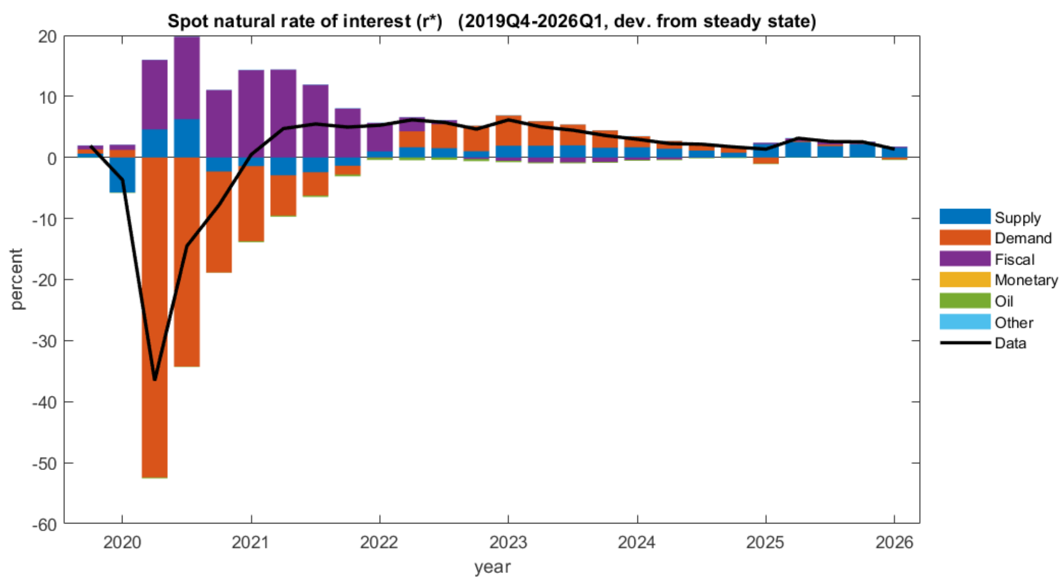

Figure 4: Spot natural rate of interest r^*: historical decomposition (paper Fig. 5b).

4.3 Detailed (12-shock) historical decompositions

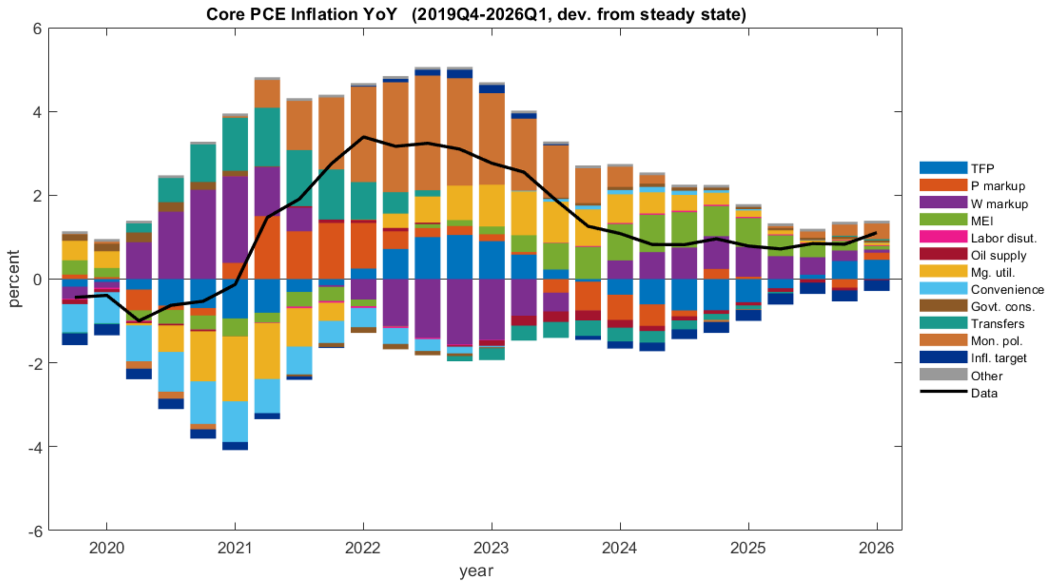

Figure 5: Detailed decomposition, YoY core PCE inflation, 12 structural shocks (paper Fig. 9a).

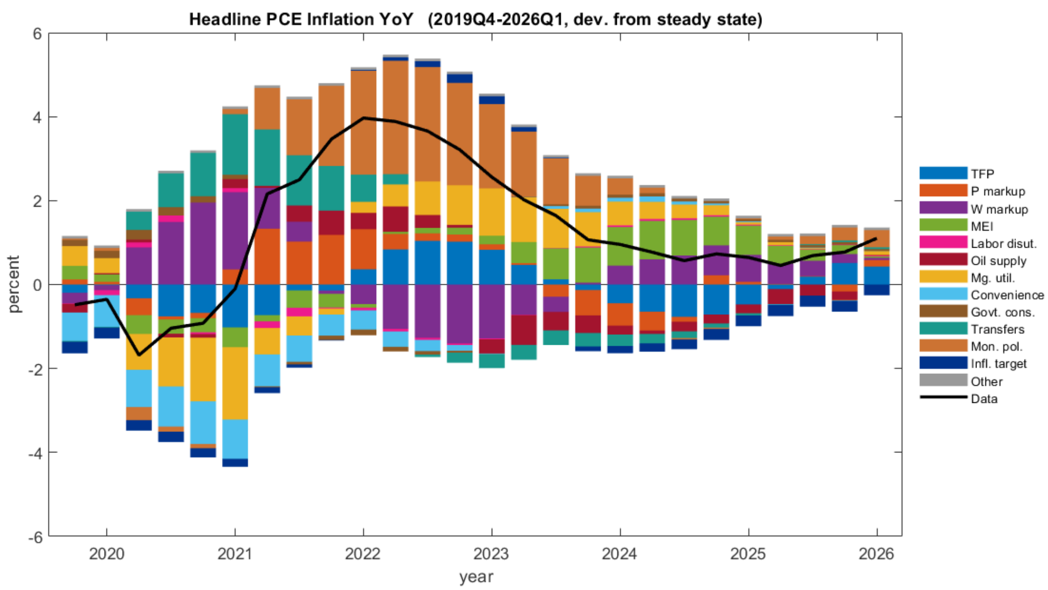

Figure 6: Detailed decomposition, YoY headline PCE inflation (paper Fig. 9b).

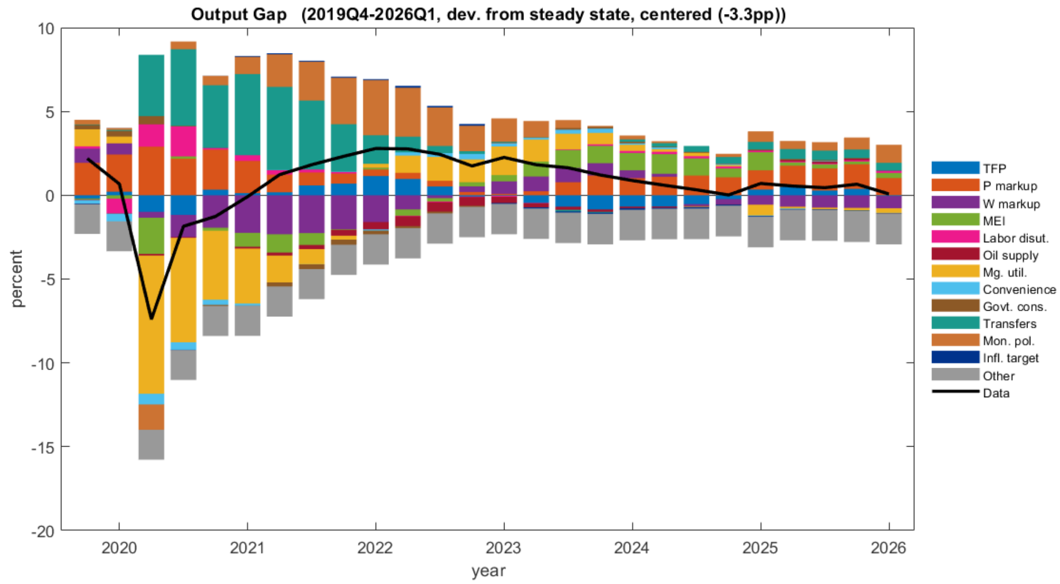

Figure 7: Detailed decomposition, output gap (paper Fig. 10a).

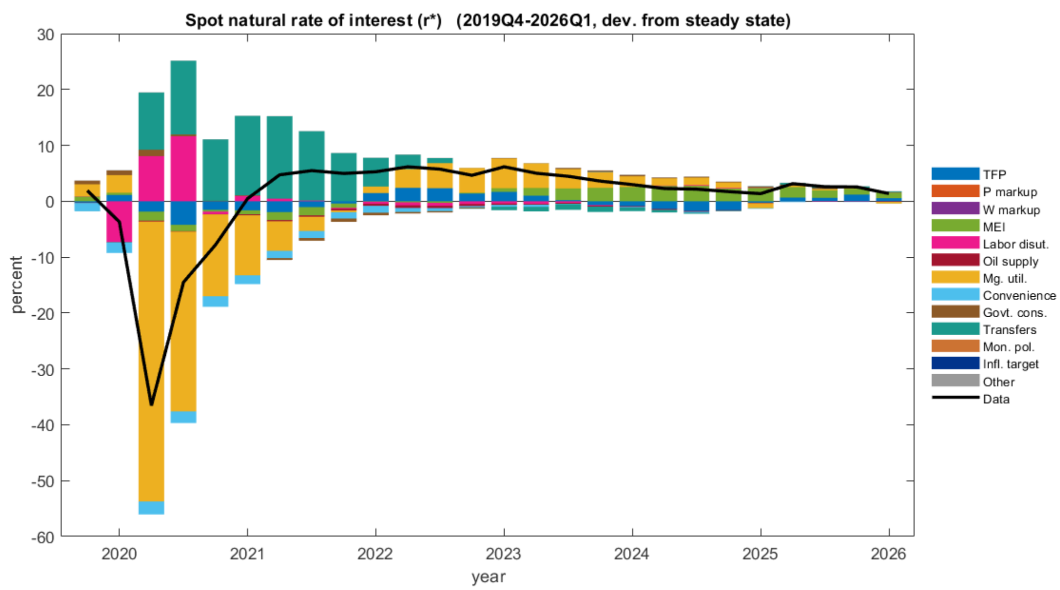

Figure 8: Detailed decomposition, spot r^* (paper Fig. 10b).

4.4 Forecast-error variance decompositions

The FEVD (Fig. 9) is computed directly from the model's order-one decision rule at the posterior point estimate and aggregated into the five shock groups. Consistent with the historical decompositions, the output gap and the natural rate are demand-dominated at business-cycle horizons; inflation is supply/markup-driven in the near term with the monetary (inflation-target) share rising at long horizons; and investment is dominated by supply/investment-specific shocks. These patterns match the paper's narrative and provide an independent cross-check on the historical decompositions.

Actual and natural output are driven by different things. The five-group aggregation pools technology and markup shocks into "supply," which obscures the flexible-price block: the constant-markup convention means ε^{μ^p} and ε^{μ^w} cannot reach GDP^f at all, so any comparison of supply shares across the two economies is partly a restatement of the convention. Splitting the group is more informative. At the one-quarter horizon the *demand* share is almost identical across the two economies, 73.4% for GDP against 71.7% for GDP^f, and the markup share of actual output is small (0.7%, rising to 35.9% unconditionally as the near-unit-root price markup accumulates). What separates them is technology and fiscal policy: technology accounts for 19.6% of actual output but only 0.4% of natural output, while fiscal shocks account for 4.4% of actual output and 27.9% of natural output, rising to 65.4% unconditionally. The technology asymmetry follows from detrending. Both series are divided by the same stochastic trend $\bar{\Gamma}$, and the flexible-price allocation tracks that trend almost one-for-one, so its detrended deviation is small; the sticky economy adjusts slowly and retains a sizeable transitory deviation. The practical implication is that r^* and the output gap in this model are preference and fiscal objects far more than technology objects, which is worth keeping in mind when reading Figures 13, 7 and 8.

The two output series share their shocks but not their propagation. $\log GDP^f$ has 98% of the standard deviation of $\log GDP$ and, as just noted, nearly the same dominant shock group at impact, yet their Hodrick–Prescott cyclical components ($\lambda = 1600$) correlate only 0.09. Identical drivers with different propagation are enough to produce that: the sticky economy filters every shock through Rotemberg pricing and the policy rule, while the flexible-price economy responds immediately. The low correlation is a statement about propagation, not evidence that the flexible-price block is mis-specified. Neither of the two obvious alternatives is a better summary: the correlation in levels (0.72) is contaminated by the near-unit-root markup wedge discussed in Section 5, and the correlation in first differences (0.04) simply over-differences two series that are already stationary.

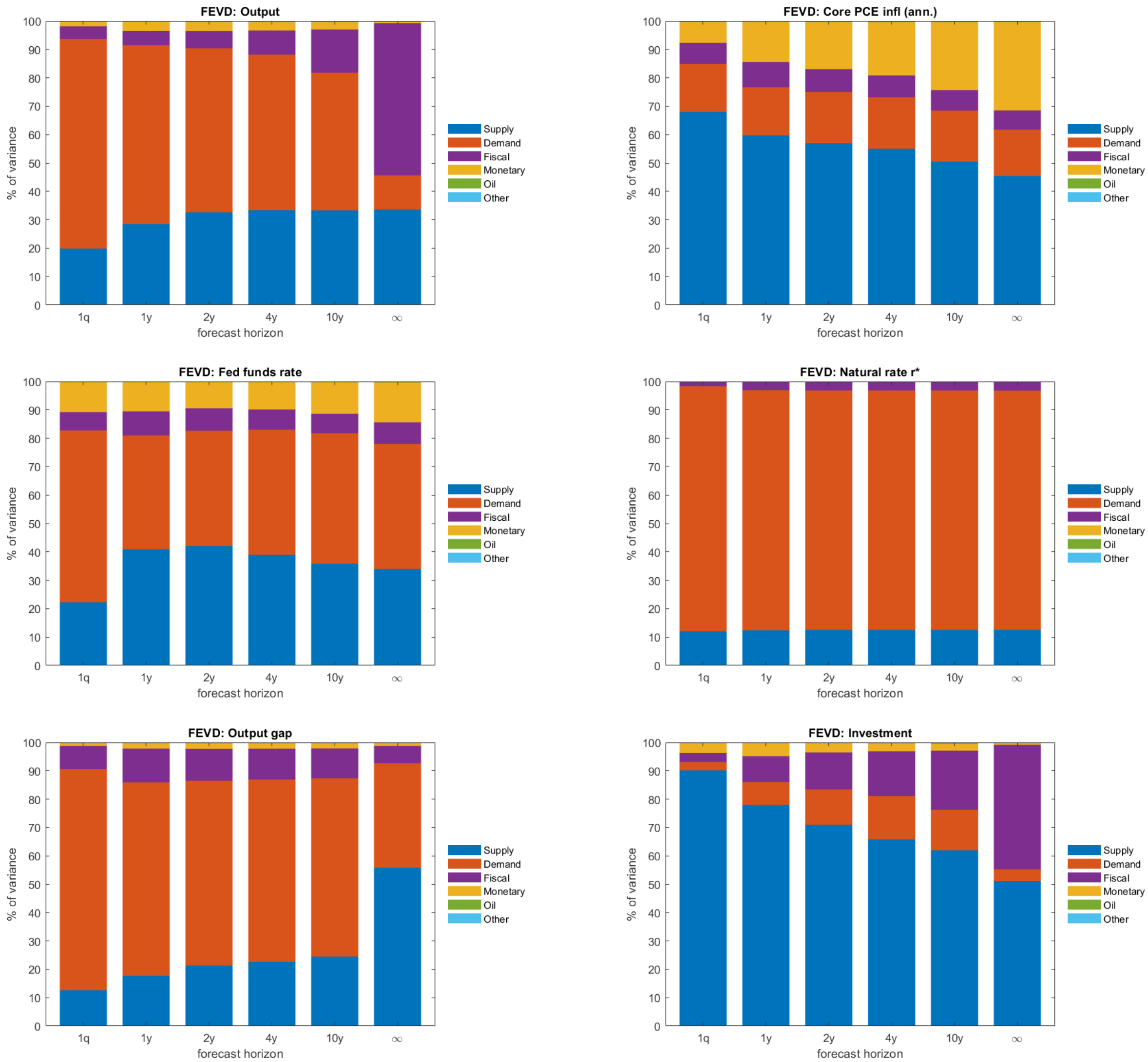

Figure 9: Forecast-error variance decompositions by horizon (output, core inflation, funds rate, r^*, output gap, investment).

4.5 Impulse responses

Figures 10–11 give the impulse responses of output growth, core and headline inflation, and the funds rate to a one-standard-deviation innovation in each of the 12 structural shocks (the paper's Appendix A), arranged with shocks in columns. The shapes and signs reproduce the paper. A useful internal check: a monetary-policy tightening leaves the natural rate essentially unchanged (the counterfactual in Fig. 12), confirming that the flexible-price block correctly isolates r^* from nominal disturbances.

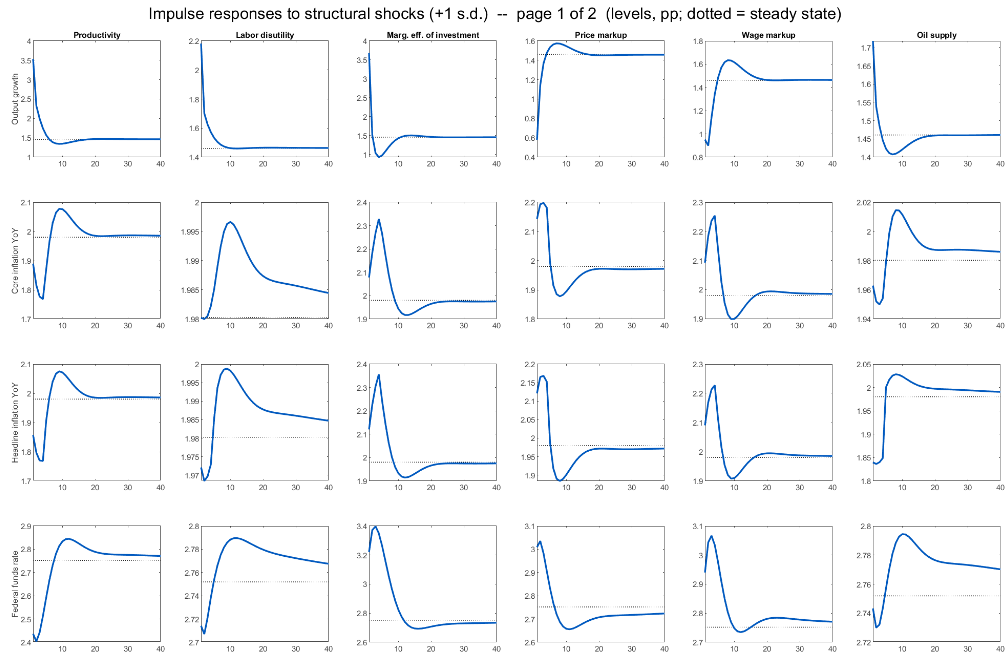

Figure 10: Impulse responses to structural shocks, page 1/2 (levels, pp; dotted line = steady state).

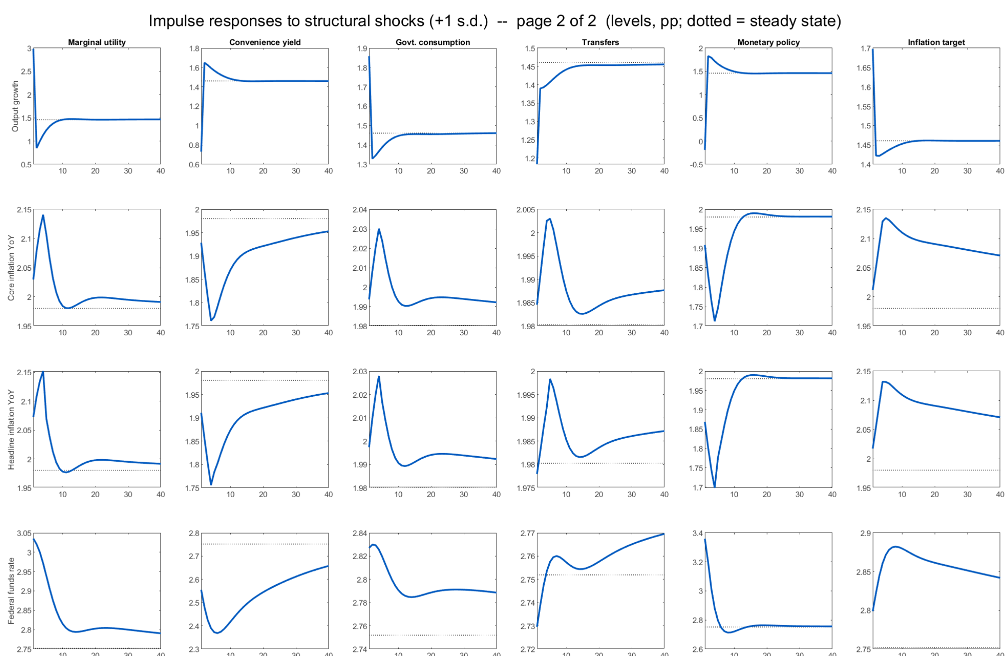

Figure 11: Impulse responses to structural shocks, page 2/2.

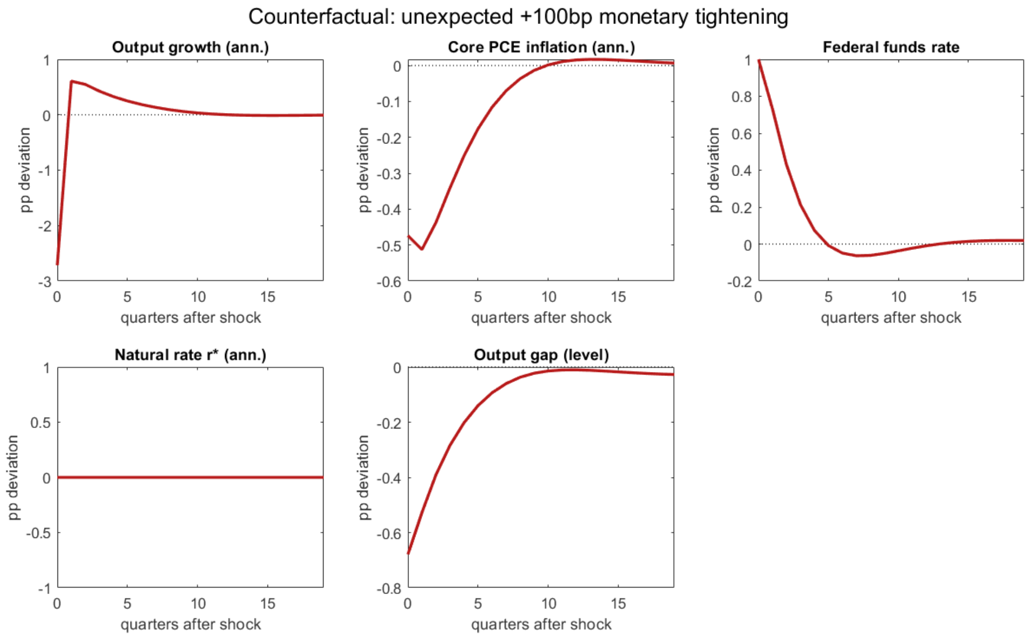

Figure 12: Counterfactual: an unexpected +100bp monetary tightening. The natural rate does not respond, as required by the monetary-neutrality of the flexible-price allocation.

4.6 The natural rate of interest

Figure 13 compares the model's one- and five-year-forward natural rates to two external measures: Laubach–Williams (one-sided, US) and Lubik–Matthes (median). As the paper reports, the model-implied r^* is broadly in line with these measures, more volatile, and somewhat lower on average, and it tracks the recent uptick in the Lubik–Matthes series.

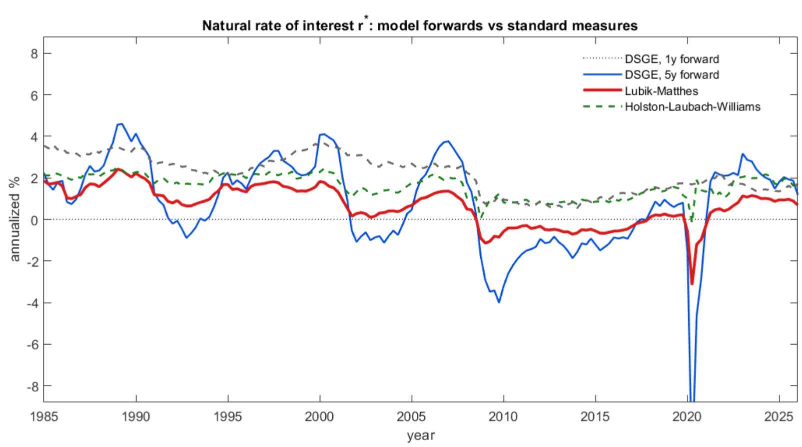

Figure 13: Model-implied r^* (1y and 5y forward) vs. Laubach–Williams (one-sided) and Lubik–Matthes (paper Fig. 6). The Laubach–Williams series is the US-only one-sided estimate from the New York Fed's current-estimates workbook, 1961Q1–2026Q1; the paper's Fig. 6 overlays the four-country Holston–Laubach–Williams measure, which differs slightly in level.

4.7 Forecasts

Figure 14 is the unconditional model forecast as of 2026Q1; Figure 15 shows two conditional scenarios (a constant-rate path and an inflation-at-target path). Table 2 reproduces the paper's Table 5 point forecasts alongside ours. The output gap matches the paper to within a tenth of a point across the horizon. The remaining differences are each attributable to one specific choice (all discussed in Section 5): (i) output growth is reported here as per-capita growth plus a ≈ 0.4pp Census population add-on, whereas the residual ≈ 0.3pp gap to the paper reflects our lower, *data-consistent* trend-growth estimate Γ^Z; (ii) core inflation reverts to the 2% target because our 10-year inflation-expectations data are anchored near 1.9%, whereas the paper's smoothed target is more persistently elevated; and (iii) the funds-rate difference is mechanically the sum of the r^* and inflation gaps, with r^* inheriting the same lower trend growth.

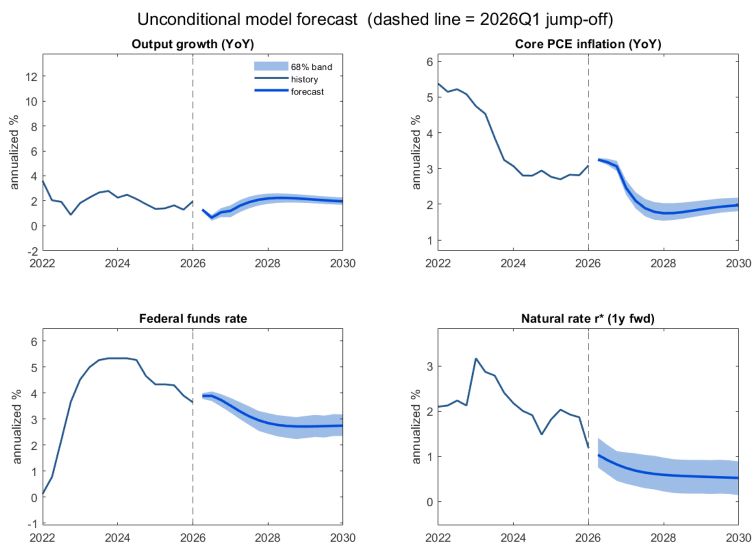

Figure 14: Unconditional model forecast, 2026Q1 jump-off (paper Fig. 7). Bands are 68% intervals.

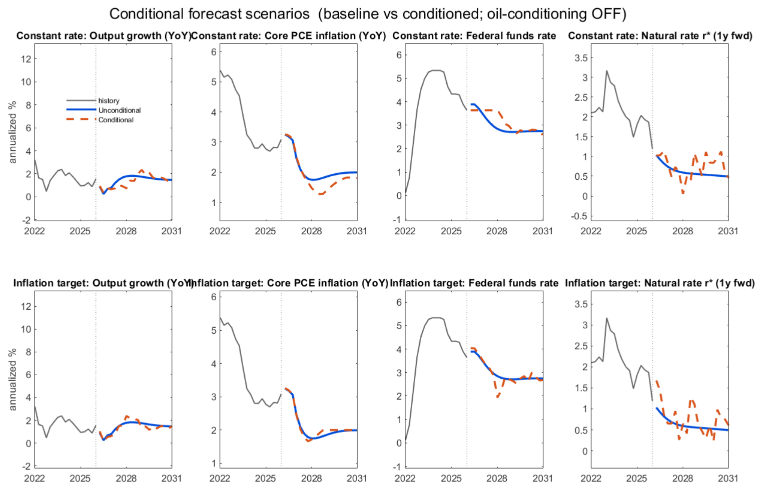

Figure 15: Conditional forecast scenarios: constant-rate and inflation-target (paper Fig. 8).

Table 2: Table 5 replication: unconditional forecast point estimates (model / paper). Output growth and core inflation are Q4/Q4; the funds rate, r^*, and the output gap are Q4 levels; "longer run" is the 2031–35 average. Output growth is aggregate (per-capita + population).

Year	Output growth model / paper	Core PCE infl. model / paper	Fed funds model / paper	r^* model / paper	Output gap model / paper
2026	1.1 / 1.7	3.1 / 3.3	3.7 / 4.0	1.1 / 0.9	−0.3 / −0.1
2027	2.1 / 2.5	1.8 / 2.5	2.9 / 3.9	0.7 / 1.0	−0.2 / −0.1
2028	2.2 / 2.5	1.8 / 2.5	2.7 / 3.9	0.6 / 1.0	0.1 / 0.1
2029	2.0 / 2.4	2.0 / 2.5	2.7 / 3.9	0.6 / 1.1	0.2 / 0.2
Longer run	1.9 / 2.2	2.0 / 2.5	2.7 / 3.8	0.4 / 1.1	0.0 / 0.1

Only output growth carries a per-capita/aggregate distinction (inflation, the funds rate, r^*, and the gap are the same object in the model and the paper). To keep Table 2 an apples-to-apples comparison without hiding the adjustment, Table 3 reports the model's *native* per-capita growth, the population add-on, the resulting aggregate growth (which is what Table 2 compares to the paper), and the paper's value. The model's per-capita and aggregate growth are both internally consistent; the aggregate column is the one comparable to the paper. The residual gap that remains *after* the population adjustment ($\approx$ 0.3–0.6pp) is *not* a population effect: it is the data-anchored trend-growth (Γ^Z) difference of Section 5. The composite trend $\bar{\Gamma} = \bar{\Gamma}^Z \bar{\Gamma}^N$, which is what per-capita growth reverts to, is 1.40% annualized on our posterior means against 1.875% on the paper's; ours tracks the realized average growth in the sample, the paper's stays at its prior.

Table 3: Output growth: model per-capita vs. aggregate (Q4/Q4, %). The model's native object is per-capita growth; aggregate growth adds a Census population-growth forecast ($\approx$ 0.4pp, the paper's footnote 3). The aggregate column is the apples-to-apples comparison with the paper.

Year	Per-capita (model)	+ Population	= Aggregate (model)	Paper (aggregate)
2026	0.7	0.4	1.1	1.7
2027	1.7	0.4	2.1	2.5
2028	1.8	0.4	2.2	2.5
2029	1.6	0.4	2.0	2.4
Longer run	1.5	0.4	1.9	2.2

5 Assumptions and things to keep in mind when replicating

This section collects the choices that materially affect a replication. Several are not stated in the paper and had to be inferred; documenting them is the main practical contribution of this note.

Build the flexible-price block; it is what generates r^* and the gap. Section 2.12 of the paper defines the flexible-price economy completely: constant markups $\bar{\mu}^p, \bar{\mu}^w$, the two Phillips curves replaced by their static counterparts, and inflation at the time-varying target $\Pi_t^f = \Pi_t^*$, with r^* given by its eq. (96). Make sure to incorporate it. Ours (a) shares the sticky economy's steady state exactly (so the gap and r^* deviations are zero at the steady state), (b) is driven by the same shocks, including the COVID special shocks, and (c) is block-recursive and therefore likelihood-neutral, which is why it is appended only at the post-estimation stage. A modeller who omits this block cannot produce Figs. 5, 6, and 10.

Why the constant-markup convention matters quantitatively. Holding the markups fixed, as Section 2.12 prescribes, is what makes the gap usable, and it is worth saying why: the estimated price-markup shock is near a random walk ($\rho_{\mu^p} \approx 0.99$), so letting it drive potential output instead makes the gap wander by tens of percentage points and flips its sign during 2020. With markups fixed the gap is well behaved and its 2020Q2 trough and post-2021 profile match the paper. The same reasoning applies to the inflation convention: because government debt is nominal and workers are non-Ricardian, inflation is not neutral in the flexible-price economy, so using $\Pi_t^f = \Pi_t^*$ rather than a constant $\bar{\Pi}$ is a substantive choice and not a normalization.

The output gap carries a slowly wandering level wedge; center it on the long run. Because the flexible-price benchmark holds markups fixed while the estimated price-markup shock is close to a random walk ($\rho_{\mu^p} = 0.9968$, a half-life of 216 quarters against a 66-year sample), that shock moves actual output but not its flexible-price counterpart, and the gap absorbs the difference as a level. The wedge averages +5.2 percentage points but drifts across decades, from +7.1 in the 1970s to +3.2 in the 2010s, so it is a slowly moving component rather than a constant. We remove it by centering the gap on its long-run (stochastic-steady-state) forecast value rather than on its sample mean. The long-run gap is then zero by construction, which matches the paper's convention, and the 2020Q2 trough lands at -8.7, inside the paper's -8 to -10 range. Centering on the sample mean instead would remove +5.2 points rather than +3.3 and shift every historical reading by nearly two percentage points, so the choice is worth stating rather than leaving implicit.

The natural rate needs no such centering because it is a rate (a growth/difference object) and does not inherit the level wedge.

Trend growth is data-anchored and sits below the paper's prior. Our posterior for TFP trend growth is $\Gamma^Z = 1.0039$ against the paper's 1.0049. The forecast reverts to the composite trend $\bar{\Gamma} = \bar{\Gamma}^Z \bar{\Gamma}^N$, so the comparable figures are $400 \log \bar{\Gamma}$: 1.40% on our posterior means against 1.875% on the paper's. Realized per-capita GDP growth averages 1.67% over 1959Q1–2026Q1 and 1.11% since 2000, so our estimate sits inside the realized range while the paper's, essentially its prior mean, lies above it. This one parameter lowers both our output-growth forecast and our r^*; to match the paper, tighten the Γ^Z and $\bar{\vartheta}$ priors, and to keep the forecast data-consistent, leave it.

The inflation forecast is pinned by the expectations data. Core inflation reverts to the 2% target ($\bar{\Pi}$) because the 10-year inflation-expectations observable is anchored near 1.9% throughout 2023–2026 (only a brief 2022 spike), while the paper's smoothed target is more persistently elevated. Both share a 2% long-run target; the difference is what the smoother reads for the *current* one, which is governed by the expectations series and the size of its measurement error. This is a data and identification choice.

The natural-rate steady state is trend growth minus a convenience-yield wedge. In steady state $\bar{r} = \bar{\Gamma}/(\beta\bar{\vartheta})$, so r^* (annualized) $\approx$ trend growth + time preference − convenience yield: sensitive to Γ^Z and $\bar{\vartheta}$, only weakly to the discount factor, which cannot move r^* on its own.

COVID shocks. Keep the COVID special shocks active in *both* the sticky and the flexible-price blocks (the paper replaces the underlying drivers everywhere). Estimate them as observables with missing values outside their windows. Getting the windows and the missing-value setup right is what prevents 2020 from inflating the ordinary shock variances.

Numerical and accounting conventions. Compile the model to C (`use_dll`) and use parallel chains; the sampler is the wall-clock bottleneck. Annualize the natural rate continuously (400 log of the gross rate) rather than by compounding ($100(g^4 - 1)$), which explodes for a volatile r^*; build forward natural rates as mean-reverting geometric-decay perpetuities. For Table 5, Q4/Q4 growth of an annualized-quarterly series equals its four-quarter average, so the two conventions coincide.

The tax treatment of the utilization cost pins down a calibration target. The cost $A(\nu_t)K_{t-1}$ is *not* deductible: utilization income sits inside the $(1 - \tau_t^k)$ bracket in (5), (10) and (38), but the cost is met out of after-tax income. The utilization condition is therefore the after-tax one, $(1 - \tau_t^k)R_t^k = A'(\nu_t)$ (eq. 12), with $A'(1) = \kappa_a = (1 - \bar{\tau}^k)\bar{R}^k$ rather than $\bar{R}^k$. Since $A(1) = 0$ the steady state is unchanged; only κ_a moves, by $(1 - \bar{\tau}^k)$, and with it the dynamics. All three places must move together, and pairing either budget convention with the wrong first-order condition leaves the block inconsistent.

6 Conclusion

The replication reproduces the model's estimation, historical decompositions, FEVD, impulse responses and natural-rate comparisons, and matches the forecast table up to a small set of differences, each traceable to one identified parameter or data input and in each case the more data-consistent value. The full derivations follow in the Appendix.

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A Extended derivations of the model equations

The remainder of this note reproduces the full derivations of the model's optimality and equilibrium conditions, from explicit sequence-form Lagrangians through detrending and the block-recursive steady state. The material keeps the paper's notation and equation numbers: boxed results tagged (P-n) reproduce the paper's equation n, and "eq. (n)" in the prose refers to the paper. The steady-state recipe referenced in Section 3 is Appendix I.

B Preliminaries and notation

The numeraire is the core (non-oil) consumption good, whose price is P_t; $\Pi_t \equiv P_t/P_{t-1}$ is gross core inflation. The nominal price of oil is P_t^O and $p_t^O \equiv P_t^O/P_t$ is the real price of oil. Both household types share the same CES consumption aggregator between core goods $C_{N,t}$ and oil goods $C_{O,t}$, with elasticity of substitution ν_C and weight ω_C. Because the aggregator appears repeatedly, we first record its gradient.

Lemma 1 (CES gradient)**.** Let

$$C_t = \left[(1-\omega_C)^{\frac{1}{\nu_C}}(C_{N,t})^{\frac{\nu_C-1}{\nu_C}} + \omega_C^{\frac{1}{\nu_C}}(C_{O,t})^{\frac{\nu_C-1}{\nu_C}}\right]^{\frac{\nu_C}{\nu_C-1}}. \tag{P-4/14}$$

Then

$$\frac{\partial C_t}{\partial C_{N,t}} = (1-\omega_C)^{\frac{1}{\nu_C}}\left(\frac{C_t}{C_{N,t}}\right)^{\frac{1}{\nu_C}}, \qquad \frac{\partial C_t}{\partial C_{O,t}} = \omega_C^{\frac{1}{\nu_C}}\left(\frac{C_t}{C_{O,t}}\right)^{\frac{1}{\nu_C}}. \tag{1}$$

Proof. Write the bracket in (P-4/14) as $\mathcal{B}_t$, so $C_t = \mathcal{B}_t^{\nu_C/(\nu_C-1)}$ and, by definition, $\mathcal{B}_t = C_t^{(\nu_C-1)/\nu_C}$. Differentiating,

$$\frac{\partial C_t}{\partial C_{N,t}} = \frac{\nu_C}{\nu_C-1}\mathcal{B}_t^{\frac{\nu_C}{\nu_C-1}-1}\cdot(1-\omega_C)^{\frac{1}{\nu_C}}\frac{\nu_C-1}{\nu_C}(C_{N,t})^{\frac{\nu_C-1}{\nu_C}-1} = \mathcal{B}_t^{\frac{1}{\nu_C-1}}(1-\omega_C)^{\frac{1}{\nu_C}}(C_{N,t})^{-\frac{1}{\nu_C}}.$$

Since $\mathcal{B}_t^{1/(\nu_C-1)} = C_t^{\frac{\nu_C-1}{\nu_C}\cdot\frac{1}{\nu_C-1}} = C_t^{1/\nu_C}$, the first identity follows; the second is symmetric. □

A useful corollary: the marginal utility of the *core* good under log utility $\log C_t$ is

$$\frac{\partial \log C_t}{\partial C_{N,t}} = \frac{1}{C_t}\frac{\partial C_t}{\partial C_{N,t}} = (1-\omega_C)^{\frac{1}{\nu_C}}C_t^{\frac{1}{\nu_C}-1}(C_{N,t})^{-\frac{1}{\nu_C}}. \tag{2}$$

Throughout, we solve the *sequence* form of each household problem. This is equivalent to the paper's recursive (Bellman) statements but makes every first-order condition explicit. Recall that a recursion $V_t = \chi_t U_t + \beta\,\mathbb{E}_t V_{t+1}$ unrolls to $V_0 = \mathbb{E}_0\sum_{t\geq 0}\beta^t\chi_t U_t$, so the marginal-utility shock χ_t simply multiplies period-t flow utility.

C Capitalists (eqs. 3–12)

C.1 The problem

Capitalists own all firms and physical capital, hold government debt, and choose the utilization rate of capital. In sequence form they solve

$$\max\ \mathbb{E}_0\sum_{t=0}^{\infty}\beta^t\,\chi_t\log(C_t^s) \tag{3}$$

subject to, for every t, the CES aggregator (P-4/14) (with s-superscripts), the budget constraint (5),

$$(1+\tau^c)\big[P_t C^s_{N,t} + P^O_t C^s_{O,t}\big] + P_t \frac{B^s_t}{R_t} + P_t I_t = P_{t-1}B^s_{t-1} + (1-\tau^k_t)P_t\big(\nu_t R^k_t K_{t-1} + D_t\big) - P_t A(\nu_t)K_{t-1} + P_t T_t + P^O_t O_{S,t}, \tag{P-5}$$

and the law of motion for capital (6),

$$K_t = (1-\delta)K_{t-1} + \zeta_t\big[1 - S(I_t/I_{t-1})\big]I_t. \tag{P-6}$$

The controls are $\{C^s_{N,t}, C^s_{O,t}, B^s_t, I_t, K_t, \nu_t\}$. Note I_{t-1} is a state because the adjustment cost $S(I_t/I_{t-1})$ links consecutive investment levels. Note also the tax treatment of the utilization cost: utilization income is taxed but the cost of generating it is *not* deductible, so $\nu_t R^k_t K_{t-1}$ sits *inside* the $(1-\tau^k_t)$ bracket while $A(\nu_t)K_{t-1}$ sits *outside* it. Running capital harder is paid for out of after-tax income. The same treatment carries through to Tobin's Q (10) and to the government's capital-tax base (38), and it is what puts the tax wedge into the utilization condition (12); the cost remains a *real* resource loss in the aggregate constraint (70), which is a separate matter.

C.2 Lagrangian

Attach multiplier $\beta^t\Lambda_t$ to (P-5) and $\beta^t\Xi_t$ to (P-6). Here Λ_t is the marginal utility of a period-t *nominal dollar*, and Ξ_t the marginal utility value of an extra unit of installed capital. The Lagrangian is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ \chi_t \log C^s_t + \Lambda_t \Big[P_{t-1}B^s_{t-1} + (1-\tau^k_t)P_t\big(\nu_t R^k_t K_{t-1} + D_t\big) - P_t A(\nu_t)K_{t-1} + P_t T_t + P^O_t O_{S,t} - (1+\tau^c)(P_t C^s_{N,t} + P^O_t C^s_{O,t}) - P_t \tfrac{B^s_t}{R_t} - P_t I_t \Big] + \Xi_t \Big[(1-\delta)K_{t-1} + \zeta_t(1 - S(I_t/I_{t-1}))I_t - K_t \Big] \Big\}. \tag{4}$$

C.3 Static consumption allocation ⇒ eq. (8)

The FOCs for the two consumption goods are

$$C^s_{N,t}: \quad \chi_t \frac{\partial \log C^s_t}{\partial C^s_{N,t}} = \Lambda_t(1+\tau^c)P_t, \tag{5}$$

$$C^s_{O,t}: \quad \chi_t \frac{\partial \log C^s_t}{\partial C^s_{O,t}} = \Lambda_t(1+\tau^c)P^O_t. \tag{6}$$

Using (2) and its oil analogue and dividing (6) by (5),

$$\left(\frac{\omega_C}{1-\omega_C}\right)^{\frac{1}{\nu_C}} \left(\frac{C^s_{O,t}}{C^s_{N,t}}\right)^{-\frac{1}{\nu_C}} = \frac{P^O_t}{P_t} \implies \frac{C^s_{O,t}}{C^s_{N,t}} = \frac{\omega_C}{1-\omega_C}\left(\frac{P_t}{P^O_t}\right)^{\nu_C}.$$

Hence

$$\boxed{C^s_{O,t} = \left(\frac{P_t}{P^O_t}\right)^{\nu_C} \frac{\omega_C}{1-\omega_C} C^s_{N,t}} \tag{P-8}$$

which is the paper's oil-consumption condition. (Oil is a complement when $\nu_C < 1$: a higher relative oil price lowers the oil share only weakly.)

C.4 The stochastic discount factor ⇒ eq. (7)

Define the marginal utility of one unit of the *core good* (the numeraire) as $\mu_t \equiv \Lambda_t P_t$. From (5) and (2),

$$\mu_t = \Lambda_t P_t = \frac{\chi_t}{1+\tau^c}(1-\omega_C)^{\frac{1}{\nu_C}}(C_t^s)^{\frac{1}{\nu_C}-1}(C_{N,t}^s)^{-\frac{1}{\nu_C}}. \tag{7}$$

The real (core-good) stochastic discount factor between t and $t+k$ is $m_{t,t+k}^s \equiv \beta^k \mu_{t+k}/\mu_t$. Substituting (7) and cancelling the $(1+\tau^c)$ and $(1-\omega_C)^{1/\nu_C}$ constants,

$$\boxed{m_{t,t+k}^s = \beta^k\Big(\frac{\chi_{t+k}}{\chi_t}\Big)\Big(\frac{C_{t+k}^s}{C_t^s}\Big)^{-\frac{\nu_C-1}{\nu_C}}\Big(\frac{C_{N,t+k}^s}{C_{N,t}^s}\Big)^{-\frac{1}{\nu_C}}} \tag{P-7}$$

where we used $\frac{1}{\nu_C}-1 = -\frac{\nu_C-1}{\nu_C}$. This is eq. (7).

C.5 Bond Euler equation ⇒ eq. (9)

B_t^s enters (4) at t (cost $-\Lambda_t P_t B_t^s/R_t$) and at $t+1$ (payoff $+\Lambda_{t+1}P_t B_t^s$, from the term $P_{t-1}B_{t-1}^s$ dated $t+1$). The FOC is

$$-\beta^t\frac{\Lambda_t P_t}{R_t} + \beta^{t+1}\mathbb{E}_t[\Lambda_{t+1}P_t] = 0 \implies 1 = R_t\,\beta\,\mathbb{E}_t\Big[\frac{\Lambda_{t+1}}{\Lambda_t}\Big].$$

Convert nominal to real marginal utility with $\Lambda_{t+1}/\Lambda_t = (\mu_{t+1}/\mu_t)(P_t/P_{t+1})$, so that $\beta\,\Lambda_{t+1}/\Lambda_t = m_{t,t+1}^s/\Pi_{t+1}$. The value function (3) as written has no bonds in utility, which yields $1 = R_t\,\mathbb{E}_t[m_{t,t+1}^s/\Pi_{t+1}]$; following the paper's footnote, the convenience yield ϑ_t, which can be micro-founded by a bonds-in-utility term, enters as a reduced-form multiplicative wedge on the return to government debt, exactly as in Del Negro et al. (2017). Because ϑ_t is known at t it can be taken outside the expectation, so

$$\boxed{1 = R_t\,\mathbb{E}_t\Big[\frac{m_{t,t+1}^s}{\Pi_{t+1}}\Big]\vartheta_t} \tag{P-9}$$

C.6 Capital utilization ⇒ eq. (12)

ν_t enters only the period-t budget, through after-tax utilization income $(1-\tau_t^k)P_t\nu_t R_t^k K_{t-1}$ and the utilization cost $P_t A(\nu_t)K_{t-1}$, which is met out of after-tax income. Differentiating and dividing by $\Lambda_t P_t K_{t-1}$, the tax factor multiplies marginal revenue only, and leaves

$$\boxed{(1-\tau_t^k)R_t^k = A'(\nu_t)} \tag{P-12}$$

which is eq. (12), implemented as (51) in the replication code: capital is run harder until the *after-tax* rental rate equals the marginal utilization cost. Because the cost is not deductible the tax factor does not cancel, and the condition is an after-tax one. Evaluated at the normalization $\bar\nu = 1$ it is the link behind the functional-form calibration $A'(1) = \kappa_a = (1-\bar\tau^k)\bar R^k = \bar\Gamma/\beta - (1-\delta)$ (p. 14), with $\bar R^k$ given by (18) below.

C.7 Capital and Tobin's Q $\Rightarrow$ eq. (10)

K_t enters (4) at t (through $-\Xi_t K_t$) and at $t+1$ (capital income plus the undepreciated stock in the law of motion). Its FOC is

$$-\beta^t \Xi_t + \beta^{t+1}\mathbb{E}_t\Big[\Lambda_{t+1}P_{t+1}\big((1-\tau^k_{t+1})\nu_{t+1}R^k_{t+1} - A(\nu_{t+1})\big) + \Xi_{t+1}(1-\delta)\Big] = 0.$$

Define Tobin's Q in core-good units, $Q^k_t \equiv \Xi_t/\mu_t = \Xi_t/(\Lambda_t P_t)$, so $\Xi_t = Q^k_t\mu_t$. Divide the FOC by μ_t and use $\beta\,\mu_{t+1}/\mu_t = m^s_{t,t+1}$ and $\beta\,\Xi_{t+1}/\mu_t = m^s_{t,t+1}Q^k_{t+1}$:

$$\boxed{Q^k_t = \mathbb{E}_t\, m^s_{t,t+1}\Big[(1-\tau^k_{t+1})\nu_{t+1}R^k_{t+1} - A(\nu_{t+1}) + (1-\delta)Q^k_{t+1}\Big]} \tag{P-10}$$

C.8 Investment $\Rightarrow$ eq. (11)

Write $S_t \equiv S(I_t/I_{t-1})$ and $S'_t \equiv S'(I_t/I_{t-1})$. Investment I_t enters (4) in three places: the budget cost $-\Lambda_t P_t I_t$; the current law of motion $\Xi_t\zeta_t(1-S_t)I_t$; and the next-period law of motion through I_t in the denominator of $S_{t+1} = S(I_{t+1}/I_t)$. The needed derivatives are

$$\frac{\partial}{\partial I_t}\big[(1-S_t)I_t\big] = 1 - S_t - \frac{I_t}{I_{t-1}}S'_t, \qquad \frac{\partial}{\partial I_t}\big[(1-S_{t+1})I_{t+1}\big] = \Big(\frac{I_{t+1}}{I_t}\Big)^2 S'_{t+1},$$

where the second uses $\partial(I_{t+1}/I_t)/\partial I_t = -I_{t+1}/I_t^2$. The FOC is

$$-\beta^t\Lambda_t P_t + \beta^t\Xi_t\zeta_t\Big[1 - S_t - \tfrac{I_t}{I_{t-1}}S'_t\Big] + \beta^{t+1}\mathbb{E}_t\Big[\Xi_{t+1}\zeta_{t+1}\big(\tfrac{I_{t+1}}{I_t}\big)^2 S'_{t+1}\Big] = 0.$$

Divide by $\mu_t = \Lambda_t P_t$ and use $\Xi_t/\mu_t = Q^k_t$ and $\beta\,\Xi_{t+1}/\mu_t = m^s_{t,t+1}Q^k_{t+1}$:

$$\boxed{1 - Q^k_t\zeta_t\Big[1 - S_t - \frac{I_t}{I_{t-1}}S'_t\Big] = \mathbb{E}_t\, m^s_{t,t+1}\,Q^k_{t+1}\zeta_{t+1}\Big(\frac{I_{t+1}}{I_t}\Big)^2 S'_{t+1}} \tag{P-11}$$

matching eq. (11).

D Workers (eqs. 13–18)

D.1 The problem

Workers consume, supply labor (though the *quantity* is chosen by unions, see Section E), receive transfers, and save in government bonds subject to a portfolio adjustment cost $\Psi(B^w_t)$. In sequence form,

$$\max\ \mathbb{E}_0\sum_{t=0}^{\infty}\beta^t\,\chi_t\big[\log(C^w_t) - v(N_t;\xi_t)\big] \tag{8}$$

subject to the CES aggregator (P-4/14) (with w-superscripts) and the budget constraint (15),

$$(1+\tau^c)\big[P_tC^w_{N,t} + P^O_tC^w_{O,t}\big] + P_t\frac{B^w_t + \Psi(B^w_t)}{R_t} = (1-\tau^l_t)W_tN_t + P_{t-1}B^w_{t-1} + P_tT_t + P_tF_t. \tag{P-15}$$

The worker's *own* controls are $\{C^w_{N,t}, C^w_{O,t}, B^w_t\}$; N_t is taken as given, so the labor-disutility term does not enter the worker's consumption/savings FOCs (it reappears in the union problem via the MRS). Attach $\beta^t\Lambda^w_t$ to (P-15).

D.2 Static allocation and SDF ⇒ eqs. (16)–(17)

The two consumption FOCs are identical in form to (5)–(6), so the oil allocation is

$$\boxed{C^w_{O,t} = \left(\frac{P_t}{P^O_t}\right)^{\nu_C} \frac{\omega_C}{1-\omega_C}\, C^w_{N,t}} \tag{P-17}$$

and, defining $\mu^w_t \equiv \Lambda^w_t P_t$ exactly as in (7) (with w-quantities), the worker SDF $m^w_{t,t+k} \equiv \beta^k \mu^w_{t+k}/\mu^w_t$ is

$$\boxed{m^w_{t,t+k} = \beta^k \left(\frac{\chi_{t+k}}{\chi_t}\right)\left(\frac{C^w_{t+k}}{C^w_t}\right)^{-\frac{\nu_C-1}{\nu_C}} \left(\frac{C^w_{N,t+k}}{C^w_{N,t}}\right)^{-\frac{1}{\nu_C}}} \tag{P-16}$$

D.3 Bond Euler with portfolio cost ⇒ eq. (18)

The novelty relative to the capitalist is that purchasing B^w_t costs $P_t(B^w_t + \Psi(B^w_t))/R_t$, so the marginal cost of a bond is $P_t(1+\Psi'(B^w_t))/R_t$. Bonds pay off $P_t B^w_t$ next period. The FOC is

$$-\beta^t \frac{\Lambda^w_t P_t\,(1+\Psi'(B^w_t))}{R_t} + \beta^{t+1}\mathbb{E}_t[\Lambda^w_{t+1}P_t] = 0 \implies 1 = R_t\,\beta\,\mathbb{E}_t\Big[\frac{\Lambda^w_{t+1}}{\Lambda^w_t}\Big]\frac{1}{1+\Psi'(B^w_t)}.$$

Converting to the real SDF as before and adding the convenience yield,

$$\boxed{1 = R_t\,\vartheta_t\,\mathbb{E}_t\Big[\frac{m^w_{t,t+1}}{\Pi_{t+1}}\,\frac{1}{1+\Psi'(B^w_t)}\Big]} \tag{P-18}$$

As the paper notes: with $\Psi' \equiv 0$ this collapses to the capitalist Euler (9); as $\Psi' \to \infty$ the worker cannot adjust bond holdings and is effectively hand-to-mouth (MPC = 1); a finite Ψ' delivers the intermediate case. Since $\Psi'(B^w_t)$ is known at t it can be pulled outside the expectation, giving the paper's displayed form.

E Labor unions and the wage Phillips curve (eqs. 19–26)

E.1 Labor aggregator: demand and the wage index ⇒ eqs. (22)–(23)

A competitive aggregator combines varieties $N_t(j)$ into the composite N_t via the CES technology (21), $N_t = \big[\int N_t(j)^{1/\mu^w_t}\,\mathrm{d}j\big]^{\mu^w_t}$, and solves $\max_{N_t,\{N_t(j)\}} W_t N_t - \int W_t(j)N_t(j)\,\mathrm{d}j$. Substituting the constraint and taking the FOC w.r.t. $N_t(j)$,

$$W_t\,\mu^w_t\Big[\int N_t(i)^{\frac{1}{\mu^w_t}}\,\mathrm{d}i\Big]^{\mu^w_t-1}\frac{1}{\mu^w_t}N_t(j)^{\frac{1}{\mu^w_t}-1} = W_t(j).$$

Using $\big[\int N_t(i)^{1/\mu^w_t}\,\mathrm{d}i\big]^{\mu^w_t-1} = N_t^{(\mu^w_t-1)/\mu^w_t}$ and solving,

$$\boxed{N_t(j) = \Big[\frac{W_t(j)}{W_t}\Big]^{-\frac{\mu^w_t}{\mu^w_t-1}} N_t} \tag{P-22}$$

Substituting (P-22) back into the CES aggregator and simplifying gives the implicit wage index

$$\boxed{W_t^{-\frac{1}{\mu^w_t-1}} = \int W_t(j)^{-\frac{1}{\mu^w_t-1}}\,\mathrm{d}j} \tag{P-23}$$

Define the (gross) demand elasticity $\varepsilon^w_t \equiv \mu^w_t/(\mu^w_t-1)$, so $N_t(j) = (W_t(j)/W_t)^{-\varepsilon^w_t}N_t$; we use this shorthand below.

E.2 The marginal rate of substitution ⇒ eq. (25)

The (tax-adjusted) MRS between labor variety j and the core good is

$$\mathrm{MRS}_t(j) = -\frac{1+\tau^c}{1-\tau^l_t}\frac{\partial v(N_t;\xi_t)/\partial N_t(j)}{\partial u(C^w_t)/\partial C^w_{N,t}}.$$

From the disutility (19), $v(N;\xi) = \frac{\bar{\xi}}{\xi^{1+\varphi}}\int \frac{N(j)^{1+\varphi}}{1+\varphi}\,\mathrm{d}j$, we get $\partial v/\partial N_t(j) = \frac{\bar{\xi}}{\xi_t^{1+\varphi}}N_t(j)^{\varphi}$. Using (2) for the denominator (with w-quantities), $\partial u/\partial C^w_{N,t} = (1-\omega_C)^{1/\nu_C}(C^w_t)^{1/\nu_C-1}(C^w_{N,t})^{-1/\nu_C}$, so its reciprocal is $(C^w_t)^{(\nu_C-1)/\nu_C}(C^w_{N,t})^{1/\nu_C}/(1-\omega_C)^{1/\nu_C}$. Combining,

$$\boxed{\mathrm{MRS}_t(j) = \frac{(1+\tau^c)\,\bar{\xi}\,N_t(j)^{\varphi}\,(C^w_t)^{\frac{\nu_C-1}{\nu_C}}\,(C^w_{N,t})^{\frac{1}{\nu_C}}}{(1-\tau^l_t)\,\xi_t^{1+\varphi}\,(1-\omega_C)^{\frac{1}{\nu_C}}}} \tag{P-25}$$

The key property for the wage Phillips curve is that $\mathrm{MRS}_t(j) \propto N_t(j)^{\varphi}$, i.e.

$$\frac{\partial\,\mathrm{MRS}_t(j)}{\partial N_t(j)} = \varphi\,\frac{\mathrm{MRS}_t(j)}{N_t(j)}. \tag{9}$$

E.3 The union's problem and its first-order condition

Union j sets the nominal wage $W_t(j)$ to maximize the present value of wage income net of labor disutility, subject to Rotemberg costs of changing $W_t(j)$ and to the demand curve (P-22). Its recursive payoff (24) is

$$\begin{aligned} V^u[W_{t-1}(j)] = \max_{W_t(j)}\ & \big[W_t(j) - P_t\,\mathrm{MRS}_t(j)\big]N_t(j) - P_t\frac{\eta_w}{2}N_t\Big[\frac{W_t(j)}{W_{t-1}(j)}\frac{1}{\Pi^{w,\text{index}}_t} - 1\Big]^2 \\ & + \beta\,\mathbb{E}_t\frac{V^u[W_t(j)]}{\Pi_{t+1}}, \end{aligned} \tag{P-24}$$

with $N_t(j) = (W_t(j)/W_t)^{-\varepsilon^w_t}N_t$. We differentiate (P-24) w.r.t. $W_t(j)$, treating the four sources of dependence in turn.

(i) Wage-income term. Since $\mathrm{MRS}_t(j) \propto N_t(j)^{\varphi}$ and $N_t(j)$ depends on $W_t(j)$ with $\partial N_t(j)/\partial W_t(j) = -\varepsilon^w_t N_t(j)/W_t(j)$, (9) gives $\partial\,\mathrm{MRS}_t(j)/\partial W_t(j) = -\varphi\varepsilon^w_t\,\mathrm{MRS}_t(j)/W_t(j)$. The product rule then yields

$$\begin{aligned}\frac{\partial}{\partial W_t(j)}\Big\{[W_t(j) - P_t\,\mathrm{MRS}_t(j)]N_t(j)\Big\} = {}& N_t(j)\Big[1 + \varphi\,\mathrm{MRS}_t(j)\,\varepsilon^w_t\frac{P_t}{W_t(j)}\Big] \\ & - [W_t(j) - P_t\,\mathrm{MRS}_t(j)]\,\varepsilon^w_t\frac{N_t(j)}{W_t(j)}.\end{aligned}$$

(ii) Current menu cost.

$$\frac{\partial}{\partial W_t(j)}\Big\{-P_t\tfrac{\eta_w}{2}N_t\big[\tfrac{W_t(j)}{W_{t-1}(j)\Pi^{w,\text{index}}_t} - 1\big]^2\Big\} = -\eta_w N_t\frac{P_t}{W_{t-1}(j)\Pi^{w,\text{index}}_t}\Big[\frac{W_t(j)}{W_{t-1}(j)\Pi^{w,\text{index}}_t} - 1\Big].$$

(iii) Continuation via the envelope theorem. $W_t(j)$ becomes next period's state. Since $W_{t-1}(j)$ enters V^u only through the menu cost,

$$\frac{\partial V^u[W_{t-1}(j)]}{\partial W_{t-1}(j)} = P_t\eta_w N_t\Big[\frac{W_t(j)}{W_{t-1}(j)\Pi^{w,\text{index}}_t} - 1\Big]\frac{W_t(j)}{W_{t-1}(j)^2\Pi^{w,\text{index}}_t}.$$

Shifting one period forward and discounting by β/Π_{t+1},

$$\frac{\partial}{\partial W_t(j)}\Big\{\beta\mathbb{E}_t \tfrac{V^u[W_t(j)]}{\Pi_{t+1}}\Big\} = \beta\eta_w\mathbb{E}_t\Big[N_{t+1}\frac{P_{t+1}W_{t+1}(j)}{W_t(j)^2\Pi^{w,\text{index}}_{t+1}\Pi_{t+1}}\Big(\frac{W_{t+1}(j)}{W_t(j)\Pi^{w,\text{index}}_{t+1}}-1\Big)\Big].$$

Setting (i)+(ii)+(iii)= 0 reproduces the paper's union FOC (unnumbered, p. 9):

$$\begin{aligned}
&N_t(j)\Big[1+\varphi\,\text{MRS}_t(j)\frac{\mu^w_t}{\mu^w_t-1}\frac{P_t}{W_t(j)}\Big] - \big[W_t(j)-P_t\,\text{MRS}_t(j)\big]\frac{\mu^w_t}{\mu^w_t-1}\frac{N_t(j)}{W_t(j)}\\
&-\eta_w N_t\frac{P_t}{W_{t-1}(j)\Pi^{w,\text{index}}_t}\Big[\frac{W_t(j)}{W_{t-1}(j)\Pi^{w,\text{index}}_t}-1\Big]\\
&+\beta\eta_w\mathbb{E}_t N_{t+1}\frac{P_{t+1}W_{t+1}(j)}{W_t(j)^2\Pi^{w,\text{index}}_{t+1}\Pi_{t+1}}\Big[\frac{W_{t+1}(j)}{W_t(j)\Pi^{w,\text{index}}_{t+1}}-1\Big]=0.
\end{aligned}\tag{10}$$

A note on the discounting convention: the recursion (P-24) discounts the union's continuation by β/Π_{t+1} rather than by the worker's full nominal SDF $m^w_{t,t+1}/\Pi_{t+1}$. We follow the paper's stated form verbatim, so that the FOC (10) matches the text term-for-term; this is the convention embedded in the wage Phillips curve (26).

E.4 Symmetry ⇒ the wage Phillips curve (26)

Impose the symmetric equilibrium $W_t(j) = W_t$, $N_t(j) = N_t$, $\text{MRS}_t(j) = \text{MRS}_t$, and define gross wage inflation $\Pi^w_t \equiv W_t/W_{t-1}$. Two identities simplify (10):

$$\frac{W_t}{W_{t-1}\Pi^{w,\text{index}}_t}=\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t},\qquad \frac{P_t}{W_{t-1}\Pi^{w,\text{index}}_t}=\frac{P_t}{W_t}\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t}.$$

Divide (10) by N_t. The first line becomes, writing $\varepsilon^w_t = \mu^w_t/(\mu^w_t-1)$,

$$\underbrace{1+\varphi\,\text{MRS}_t\,\varepsilon^w_t\frac{P_t}{W_t}}_{\text{term (i-a)}}\underbrace{-\varepsilon^w_t-\varepsilon^w_t\frac{P_t}{W_t}\,\text{MRS}_t}_{\text{term (i-b)}}=(1-\varepsilon^w_t)+\varepsilon^w_t\frac{P_t}{W_t}\,\text{MRS}_t(1+\varphi).$$

Now multiply the *entire* divided FOC by the real wage W_t/P_t. The first group becomes exactly

$$\frac{W_t}{P_t}(1-\varepsilon^w_t)+\varepsilon^w_t\,\text{MRS}_t(1+\varphi)=\frac{W_t}{P_t}+\frac{\mu^w_t}{\mu^w_t-1}\Big[\text{MRS}_t(1+\varphi)-\frac{W_t}{P_t}\Big],\tag{11}$$

which are the first two terms of (26). For the current menu cost, $\frac{W_t}{P_t}\times$ term (ii) $= -\eta_w\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t}\big(\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t}-1\big)$. For the continuation, use the crucial cancellation

$$\frac{W_t}{P_t}\cdot\frac{P_{t+1}W_{t+1}}{W_t^2\,\Pi^{w,\text{index}}_{t+1}\Pi_{t+1}}=\frac{P_{t+1}}{P_t\,\Pi_{t+1}}\cdot\frac{W_{t+1}}{W_t}\cdot\frac{1}{\Pi^{w,\text{index}}_{t+1}}=\frac{\Pi^w_{t+1}}{\Pi^{w,\text{index}}_{t+1}},$$

since $P_{t+1}/(P_t\Pi_{t+1}) = 1$. Collecting all four pieces gives the wage Phillips curve

$$\boxed{\frac{W_t}{P_t}+\frac{\mu^w_t}{\mu^w_t-1}\Big[\text{MRS}_t(1+\varphi)-\frac{W_t}{P_t}\Big]-\eta_w\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t}\Big(\frac{\Pi^w_t}{\Pi^{w,\text{index}}_t}-1\Big)+\beta\eta_w\mathbb{E}_t\frac{N_{t+1}}{N_t}\frac{\Pi^w_{t+1}}{\Pi^{w,\text{index}}_{t+1}}\Big(\frac{\Pi^w_{t+1}}{\Pi^{w,\text{index}}_{t+1}}-1\Big)=0}\tag{P-26}$$

which is the paper's eq. (26).

Reading (P-26). Ignoring adjustment costs ($\eta_w = 0$), the first two terms give $\frac{W_t}{P_t}(1-\varepsilon_t^w) + \varepsilon_t^w \,\mathrm{MRS}_t(1+\varphi) = 0$, i.e.

$$\frac{W_t}{P_t} = \frac{\varepsilon_t^w}{\varepsilon_t^w - 1}\,\mathrm{MRS}_t(1+\varphi) = \mu_t^w\,\mathrm{MRS}_t(1+\varphi),$$

since $\varepsilon_t^w - 1 = 1/(\mu_t^w - 1)$. The flex-wage limit therefore sets the real wage as a markup μ_t^w over the "competitive" object $\mathrm{MRS}_t(1+\varphi)$; this is the condition that reappears in the steady state as (26). The two inflation terms are the standard forward-looking Rotemberg wedge, indexed to $\Pi_t^{w,\mathrm{index}}$.

F Goods producers (eqs. 27–36)

Production has the same two-layer structure as labor supply: competitive *final* core-good producers aggregate a continuum of varieties k, and monopolistically competitive *intermediate* producers set prices subject to Rotemberg costs. We take the two layers in turn.

F.1 Final core-good producers: demand and price index ⇒ eq. (27)

The representative final producer solves, exactly as the labor aggregator,

$$\max_{Y_t,\{Y_t(k)\}} P_tY_t - \int P_t(k)Y_t(k)\,\mathrm{d}k \qquad \text{s.t.} \qquad Y_t = \left[\int Y_t(k)^{\frac{1}{\mu_t^p}}\,\mathrm{d}k\right]^{\mu_t^p}.$$

Substituting the constraint and taking the FOC w.r.t. $Y_t(k)$ (identical algebra to (P-22)) gives the variety demand curve and, from the zero-profit condition, the price index

$$\boxed{Y_t(k) = \left[\frac{P_t(k)}{P_t}\right]^{-\frac{\mu_t^p}{\mu_t^p-1}} Y_t}, \qquad \boxed{P_t^{-\frac{1}{\mu_t^p-1}} = \int P_t(k)^{-\frac{1}{\mu_t^p-1}}\,\mathrm{d}k} \tag{P-27}$$

Define the (gross) demand elasticity $\varepsilon_t^p \equiv \mu_t^p/(\mu_t^p-1)$, so $Y_t(k) = (P_t(k)/P_t)^{-\varepsilon_t^p}Y_t$.

F.2 Intermediate producers: cost minimization ⇒ eqs. (32)–(35)

An intermediate producer of variety k combines a non-oil input $V_t(k)$ and oil $O_{Y,t}(k)$ through a CES technology (30) into output, and produces the non-oil input via Cobb–Douglas over utilized capital and efficiency labor (31),

$$Y_t(k) \le \left[(1-\omega_Y)^{\frac{1}{\nu_Y}} V_t(k)^{\frac{\nu_Y-1}{\nu_Y}} + \omega_Y^{\frac{1}{\nu_Y}} O_{Y,t}(k)^{\frac{\nu_Y-1}{\nu_Y}}\right]^{\frac{\nu_Y}{\nu_Y-1}}, \qquad V_t(k) = \big(K_t^u(k)\big)^{\alpha}\big(Z_tN_t(k)\big)^{1-\alpha}. \tag{12}$$

Cost minimization is done in two nested layers.

Layer 1: capital vs. labor ⇒ (32)–(33). Minimizing $W_tN_t(k) + P_tR_t^kK_t^u(k)$ subject to a given $V_t(k)$, the FOCs equate each marginal product (times the multiplier) to the input's price. Using $\partial V/\partial K^u = \alpha V/K^u$ and $\partial V/\partial N = (1-\alpha)V/N$ and taking the ratio,

$$\boxed{K_t^u(k) = \frac{\alpha}{1-\alpha}\frac{W_t}{P_tR_t^k}\,N_t(k)} \tag{P-32}$$

Substituting (P-32) back into the production function gives $N_t(k) = V_t(k)\, Z_t^{-(1-\alpha)} \big(\frac{1-\alpha}{\alpha} \frac{P_t R_t^k}{W_t}\big)^{\alpha}$, and total cost collapses to $W_t N_t(k)/(1-\alpha)$ because $W_t N + P_t R_t^k K^u = W_t N\big(1 + \frac{\alpha}{1-\alpha}\big)$. The (per-unit) nominal marginal cost of the non-oil input is therefore

$$\boxed{P_t^V = Z_t^{-(1-\alpha)} \Big(\frac{P_t R_t^k}{\alpha}\Big)^{\alpha} \Big(\frac{W_t}{1-\alpha}\Big)^{1-\alpha}} \tag{P-33}$$

which is independent of k (all firms face the same factor prices), so $P_t^V(k) = P_t^V$.

Layer 2: non-oil input vs. oil ⇒ (34)–(35). Now minimize $P_t^V V_t(k) + P_t^O O_{Y,t}(k)$ subject to the CES aggregator (12). This is structurally identical to the household oil-allocation problem, so the input mix is

$$\boxed{V_t(k) = \Big(\frac{1-\omega_Y}{\omega_Y}\Big)\Big(\frac{P_t^O}{P_t^V}\Big)^{\nu_Y} O_{Y,t}(k)} \tag{P-34}$$

For the marginal cost, substitute the conditional factor demands $V = Y(1-\omega_Y)(P_t MC_t/P_t^V)^{\nu_Y}$ and $O_Y = Y\omega_Y(P_t MC_t/P_t^O)^{\nu_Y}$ (where the multiplier $P_t MC_t$ is the marginal cost) into total cost $P_t^V V + P_t^O O_Y = P_t MC_t\, Y$. Cancelling Y and using constant returns yields the CES cost dual

$$\boxed{P_t MC_t = \Big[\omega_Y (P_t^O)^{1-\nu_Y} + (1-\omega_Y)(P_t^V)^{1-\nu_Y}\Big]^{\frac{1}{1-\nu_Y}}} \tag{P-35}$$

which is eq. (35). Marginal cost is common across k.

F.3 Pricing: the price NKPC ⇒ eq. (36)

With marginal cost in hand, the producer's only remaining choice is the nominal price $P_t(k)$, set to maximize the present value of nominal profits net of Rotemberg costs. Firms are owned by capitalists, so they discount *nominal* payoffs with the capitalists' *nominal* SDF,

$$M_{t,t+1} \equiv \frac{m^s_{t,t+1}}{\Pi_{t+1}} = \beta\, \frac{\mu_{t+1}}{\mu_t}\, \frac{P_t}{P_{t+1}}. \tag{13}$$

Pricing the nominal claim with the nominal kernel matters for the algebra below: the $1/\Pi_{t+1}$ in $M_{t,t+1}$ is exactly what cancels one power of Π_{t+1} in the forward term and delivers the NKPC (36)/(62). Equivalently, one may deflate the whole problem and discount *real* profits by $m^s_{t,t+1}$; we use the nominal kernel (13) throughout. The pricing problem is

$$V^f[P_{t-1}(k)] = \max_{P_t(k)}\, [P_t(k) - P_t MC_t] Y_t(k) - P_t \frac{\eta_p}{2} Y_t \Big[\frac{P_t(k)}{P_{t-1}(k)\Pi_t^{p,\text{index}}} - 1\Big]^2 + \mathbb{E}_t M_{t,t+1} V^f[P_t(k)],$$

with $Y_t(k) = (P_t(k)/P_t)^{-\varepsilon_t^p} Y_t$. The three pieces of the FOC mirror the union problem of Section E exactly.

(i) Sales term. $\dfrac{\partial}{\partial P_t(k)}\big\{[P_t(k) - P_t MC_t] Y_t(k)\big\} = Y_t(k)\Big[1 - \varepsilon_t^p + \varepsilon_t^p \dfrac{P_t MC_t}{P_t(k)}\Big]$, using $\partial Y_t(k)/\partial P_t(k) = -\varepsilon_t^p Y_t(k)/P_t(k)$.

(ii) Current menu cost. $-\eta_p P_t Y_t \dfrac{1}{P_{t-1}(k)\Pi_t^{p,\text{index}}}\Big[\dfrac{P_t(k)}{P_{t-1}(k)\Pi_t^{p,\text{index}}} - 1\Big].$

(iii) Continuation (envelope). Since $P_{t-1}(k)$ enters V^f only through the menu cost, $\partial V^f[P_t(k)]/\partial P_t(k) = \eta_p P_{t+1} Y_{t+1} \big[\frac{P_{t+1}(k)}{P_t(k)\Pi^{p,\text{index}}_{t+1}} - 1\big] \frac{P_{t+1}(k)}{P_t(k)^2 \Pi^{p,\text{index}}_{t+1}}$, discounted by $M_{t,t+1}$.

Setting (i)+(ii)+(iii)= 0, imposing symmetry $P_t(k) = P_t$, $Y_t(k) = Y_t$, and writing $\Pi_t = P_t/P_{t-1}$, the current menu term becomes $-\eta_p(\Pi_t/\Pi^{p,\text{index}}_t)(\Pi_t/\Pi^{p,\text{index}}_t - 1)$, while the continuation, after the $1/\Pi_{t+1}$ in $M_{t,t+1}$ cancels one power of Π_{t+1}, becomes $\mathbb{E}_t m^s_{t,t+1} \eta_p (Y_{t+1}/Y_t)(\Pi_{t+1}/\Pi^{p,\text{index}}_{t+1})(\Pi_{t+1}/\Pi^{p,\text{index}}_{t+1} - 1)$. Dividing the sales term by Y_t and using $1 - \varepsilon^p_t + \varepsilon^p_t MC_t = (\mu^p_t MC_t - 1)/(\mu^p_t - 1)$ (since $\varepsilon^p_t = \mu^p_t/(\mu^p_t - 1)$), the FOC rearranges to the New Keynesian Phillips curve

$$\boxed{\frac{\Pi_t}{\Pi^{p,\text{index}}_t}\Big(\frac{\Pi_t}{\Pi^{p,\text{index}}_t} - 1\Big) = \mathbb{E}_t\Big[m^s_{t,t+1}\frac{Y_{t+1}}{Y_t}\frac{\Pi_{t+1}}{\Pi^{p,\text{index}}_{t+1}}\Big(\frac{\Pi_{t+1}}{\Pi^{p,\text{index}}_{t+1}} - 1\Big)\Big] + \frac{1}{\eta_p(\mu^p_t - 1)}\big(\mu^p_t MC_t - 1\big)} \tag{P-36}$$

which is eq. (36). In the flexible-price limit ($\eta_p \to 0$, i.e. adjustment costs removed) the last term forces $\mu^p_t MC_t = 1$: price equals the markup μ^p_t over marginal cost, matching the paper's flexible-price condition $\bar{\mu}^p MC_t = 1$.

F.4 Firm profits

Nominal profit of variety k is revenue minus factor payments and menu costs. Integrating over the symmetric equilibrium ($P_t(k) = P_t$, $Y_t(k) = Y_t$, $\int N_t(k)\,\mathrm{d}k = N_t$, etc.), the total profit rebated to capitalists is

$$\boxed{P_t D_t = P_t Y_t - W_t N_t - P_t R^k_t K^u_t - P^O_t O_{Y,t} - P_t \frac{\eta_p}{2} Y_t \Big(\frac{\Pi_t}{\Pi^{p,\text{index}}_t} - 1\Big)^2} \tag{P-(p.11)}$$

with $K^u_t = \nu_t K_{t-1}$. This is the paper's (unnumbered) profit identity on p. 11 and feeds the capitalists' budget constraint (P-5).

G Oil market, government, and aggregation (eqs. 37–43, 70)

G.1 Oil market clearing ⇒ eq. (37)

Capitalists own a stochastic oil endowment $O_{S,t} \equiv o_{S,t} Z_t \xi_t$ (net global supply, riding the balanced-growth trend). The economy is closed, so all oil is used domestically, either consumed by households or burned as an intermediate input. With aggregate household oil consumption $C_{O,t} \equiv \lambda C^w_{O,t} + (1 - \lambda) C^s_{O,t}$ and firm oil demand $O_{Y,t}$, market clearing is

$$\boxed{C_{O,t} + O_{Y,t} = O_{S,t}} \tag{P-37}$$

and the real oil price $p^O_t = P^O_t/P_t$ adjusts to clear it. (This closed-economy assumption is what lets the oil terms net out of the resource constraint below.)

G.2 Government budget constraint ⇒ eq. (38)

The fiscal authority's budget is an accounting identity equating uses of funds (spending, transfers, and repaying maturing debt) to sources (tax revenue and new debt issuance):

$$\boxed{P_t G_t + P_t T_t + P_{t-1} B^g_{t-1} = \tau^c(P_t C_{N,t} + P^O_t C_{O,t}) + \tau^l_t W_t N_t + \tau^k_t P_t\big(\nu_t R^k_t K_{t-1} + D_t\big) + P_t \frac{B^g_t}{R_t}} \tag{P-38}$$

with $C_{j,t} = \lambda C^w_{j,t} + (1-\lambda)C^s_{j,t}$ for $j = N, O$. New debt B^g_t is issued at the discount price $1/R_t$; maturing debt B^g_{t-1} is repaid at the previous price level P_{t-1}. The consumption-tax base spans both core and oil expenditure; labor income is taxed at τ^l_t and capital income plus profits at τ^k_t, with no deduction for the utilization cost. The base must match the capitalists' budget (P-5) exactly: if $A(\nu_t)K_{t-1}$ were deducted in one and not the other, the τ^k_t terms would fail to cancel when the budgets are summed and the resource constraint (P-70) would be off by $\tau^k_t A(\nu_t)K_{t-1}$.

G.3 Fiscal and monetary rules ⇒ eqs. (39)–(41)

These are *postulated behavioral rules*, not the outcome of an optimization, so there is nothing to "derive", but it is worth recording their common structure. Each labor- and capital-tax rate partially adjusts toward a target that rises with the debt-to-GDP gap, while the policy rate follows a smoothed Taylor rule in inflation and the GDP gap:

$$\tau^l_t = (\tau^l_{t-1})^{\rho_{\tau,l}} \Big[\bar{\tau}^l \big(\tfrac{B^g_{t-1}/GDP_{t-1}}{\bar{B}^g/GDP}\big)^{\phi_{\tau,l}}\Big]^{1-\rho_{\tau,l}}, \tag{P-39}$$

$$\tau^k_t = (\tau^k_{t-1})^{\rho_{\tau,k}} \Big[\bar{\tau}^k \big(\tfrac{B^g_{t-1}/GDP_{t-1}}{\bar{B}^g/GDP}\big)^{\phi_{\tau,k}}\Big]^{1-\rho_{\tau,k}}, \tag{P-40}$$

$$R_t = R^{\rho_r}_{t-1} \Big[\bar{r}\,\Pi^*_t \big(\tfrac{\Pi_t}{\Pi^*_t}\big)^{\phi_\Pi} \big(\tfrac{GDP_t}{\Gamma_t GDP_{t-1}}\big)^{\phi_Y}\Big]^{1-\rho_r} \mathrm{mp}_t. \tag{P-41}$$

The exponents ρ control inertia; $\phi_{\tau,x}$ set how fast taxes chase the debt ratio (a low $\phi_{\tau,x}$ makes debt the residual financing instrument, so fiscal expansions are deficit-financed); ϕ_Π, ϕ_Y are the Taylor coefficients; and mp_t is the monetary-policy shock. The consumption tax τ^c is constant, and is pinned down by the steady-state government budget ((28) below) rather than calibrated directly. The two tax rules are written with separate persistences for generality; the implementation imposes the common value $\rho_{\tau,l} = \rho_{\tau,k} = \rho_\tau$. Note the Taylor rule strips out trend growth Γ_t so it reacts only to the cyclical component of GDP.

G.4 Aggregate resource constraint ⇒ eq. (70)

This is the one genuinely nontrivial aggregation. Sum the worker budget (P-15), the capitalist budget (P-5), and the government budget (P-38), using aggregate consumption $C_{j,t} = \lambda C^w_{j,t} + (1-\lambda)C^s_{j,t}$. Four equilibrium conditions make almost everything cancel:

(i) **Bond market clearing:** household holdings equal government debt, $\lambda B^w_t + (1-\lambda)B^s_t = B^g_t$, so both the B_t/R_t (issuance) and B_{t-1} (repayment) terms cancel between households and government.

(ii) **Portfolio-cost rebate:** F_t is set to return the adjustment cost, $F_t = \Psi(B^w_t)/R_t$, so the Ψ term drops out (no resource is lost to portfolio frictions).

(iii) **Transfers and taxes:** households receive T_t that the government pays; the consumption-, labor-, and capital-tax revenues on the government's source side cancel the corresponding τ^c, τ^l, τ^k wedges on the households' use/source sides, leaving pre-tax quantities.

(iv) **Firm profits:** substitute the profit identity (P-(p.11)).

After the cancellations in (i)–(iii), the summed constraint reads

$$P_t C_{N,t} + P^O_t C_{O,t} + P_t I_t + P_t G_t = W_t N_t + P_t \nu_t R^k_t K_{t-1} + P_t D_t - P_t A(\nu_t) K_{t-1} + P^O_t O_{S,t}. \tag{14}$$

Now insert $P_t D_t$ from (P-(p.11)) with $K^u_t = \nu_t K_{t-1}$: the wage bill $W_t N_t$ and the capital-rental bill $P_t \nu_t R^k_t K_{t-1}$ cancel, leaving

$$P_t C_{N,t} + P^O_t C_{O,t} + P_t I_t + P_t G_t = P_t Y_t - P^O_t O_{Y,t} - P_t \tfrac{\eta_p}{2} Y_t \big(\tfrac{\Pi_t}{\Pi^{p,\mathrm{index}}_t} - 1\big)^2 - P_t A(\nu_t) K_{t-1} + P^O_t O_{S,t}.$$

Finally apply oil market clearing (P-37): $P_t^O(O_{S,t} - O_{Y,t}) = P_t^O C_{O,t}$ cancels the $P_t^O C_{O,t}$ on the left. Dividing by P_t gives the core-goods resource constraint

$$\boxed{C_{N,t} + G_t + I_t = Y_t\Big[1 - \frac{\eta_p}{2}\Big(\frac{\Pi_t}{\Pi_t^{p,\text{index}}} - 1\Big)^2\Big] - A(\nu_t)K_{t-1}} \tag{P-70}$$

Core output is absorbed by core consumption, government purchases, and investment, minus the resources lost to price-adjustment and capital-utilization costs. *All oil terms have vanished*: because the economy is closed and the endowment is domestically owned, oil is a pure intermediate/consumption wash in the core-good market. (The paper's eq. (70) is the detrended counterpart of (P-70).)

G.5 GDP and headline inflation ⇒ eqs. (42)–(43)

GDP. Model GDP is core output plus the value of oil *net* of the oil used up as an intermediate input (to avoid double-counting the latter):

$$\boxed{GDP_t = Y_t + p_t^O\big(O_{S,t} - O_{Y,t}\big)} \tag{P-42}$$

Using (P-37), $O_{S,t} - O_{Y,t} = C_{O,t}$, so $GDP_t = Y_t + p_t^O C_{O,t}$: value added equals core output plus final oil consumption.

Headline inflation. The consumption-based price index dual to the CES aggregator (P-4/14), the minimum core-good expenditure buying one unit of C_t, is $P_t^c = \big[(1-\omega_C)P_t^{1-\nu_C} + \omega_C(P_t^O)^{1-\nu_C}\big]^{1/(1-\nu_C)}$ (same dual as (P-35)). Headline inflation is $\Pi_t^h \equiv P_t^c/P_{t-1}^c$. Factoring $P_t^{1-\nu_C}$ from the numerator and $P_{t-1}^{1-\nu_C}$ from the denominator and using $p_t^O = P_t^O/P_t$,

$$\boxed{\Pi_t^h = \Pi_t \times \Big[\frac{1 - \omega_C + \omega_C(p_t^O)^{1-\nu_C}}{1 - \omega_C + \omega_C(p_{t-1}^O)^{1-\nu_C}}\Big]^{\frac{1}{1-\nu_C}}} \tag{P-43}$$

i.e. core inflation Π_t scaled by the change in the relative-oil-price adjustment of the consumption basket. This is the model's counterpart to headline (vs. core) PCE inflation.

H Detrending: from levels to the stationary system (eqs. 44–95)

The model has two stochastic trends: labor-augmenting TFP Z_t (growth Γ_t^Z) and the labor disutility shifter ξ_t (growth Γ_t^N). Most real quantities inherit the composite trend $Z_t\xi_t$, with gross growth $\Gamma_t \equiv \Gamma_t^Z \Gamma_t^N$. To obtain a stationary system we divide each variable by its trend.

H.1 The change of variables

For a generic real quantity X_t (consumption, investment, capital, output, government spending, transfers, real bonds), set

$$x_t \equiv \frac{X_t}{Z_t\xi_t}, \qquad n_t \equiv \frac{N_t}{\xi_t} \text{ (hours)}, \qquad w_t \equiv \frac{W_t/P_t}{Z_t} \text{ (real wage)}. \tag{15}$$

Hours are scaled by ξ_t only and the real wage by Z_t only; this keeps the real wage bill $\frac{W_t}{P_t}N_t = w_t n_t Z_t\xi_t$ on the composite trend. Prices, gross rates, ratios, the utilization rate, Tobin's Q, real marginal cost, the markups, and the tax rates are already stationary. The real oil price $p^O_t = P^O_t/P_t$ is stationary (the oil endowment $O_{S,t} = o_{S,t} Z_t \xi_t$ rides the trend, and $o_{S,t}$ is stationary). Two facts are used repeatedly:

$$\frac{X_{t+1}}{X_t} = \frac{x_{t+1}}{x_t}\Gamma_{t+1}, \qquad \frac{X_{t-1}}{Z_t\xi_t} = \frac{x_{t-1}}{\Gamma_t}, \qquad \frac{P_{t-1}}{P_t} = \frac{1}{\Pi_t}. \tag{16}$$

We show the trend algebra on the conditions where it actually bites; the rest follow by the same substitution.

H.2 Where the trend bites: worked cases

Stochastic discount factors (44)/(49). In $m^s_{t,t+1} = \beta(\frac{\chi_{t+1}}{\chi_t})(\frac{C^s_{t+1}}{C^s_t})^{-\frac{\nu_C-1}{\nu_C}}(\frac{C^s_{N,t+1}}{C^s_{N,t}})^{-\frac{1}{\nu_C}}$, each consumption ratio carries a factor Γ_{t+1} by (16). The two trend exponents sum to $-\frac{\nu_C-1}{\nu_C} - \frac{1}{\nu_C} = -1$, so exactly one power of Γ_{t+1} survives:

$$\boxed{m^s_{t,t+1} = \frac{\beta}{\Gamma_{t+1}}\left(\frac{\chi_{t+1}}{\chi_t}\right)\left(\frac{c^s_{t+1}}{c^s_t}\right)^{-\frac{\nu_C-1}{\nu_C}}\left(\frac{c^s_{N,t+1}}{c^s_{N,t}}\right)^{-\frac{1}{\nu_C}}} \tag{P-49}$$

and identically for the worker SDF (44). This β/Γ_{t+1} is the reason trend growth enters the steady-state real interest rate below.

Worker budget constraint (46). Divide the levels budget (P-15) by $P_t Z_t \xi_t$. Consumption terms give $(1+\tau^c)(c^w_{N,t} + p^O_t c^w_{O,t})$; the new-bond term $\frac{P_t B^w_t}{R_t P_t Z_t \xi_t} = \frac{b^w_t}{R_t}$; labor income $(1-\tau^l_t)\frac{W_t}{P_t}\frac{N_t}{Z_t\xi_t} = (1-\tau^l_t) w_t n_t$, which written *per worker* (there are λ of them, and n_t is aggregate) is $(1-\tau^l_t)w_t n_t/\lambda$; the maturing-bond term $\frac{P_{t-1}B^w_{t-1}}{P_t Z_t \xi_t} = \frac{b^w_{t-1}}{\Pi_t\Gamma_t}$ by (16). The portfolio cost Ψ and rebate F_t cancel (Section G), so

$$\boxed{(1+\tau^c)(c^w_{N,t} + p^O_t c^w_{O,t}) + \frac{b^w_t}{R_t} = (1-\tau^l_t)\frac{w_t n_t}{\lambda} + \frac{b^w_{t-1}}{\Pi_t\Gamma_t} + t_t} \tag{P-46}$$

Capital law of motion (60) and utilized capital (61). Dividing (P-6) by $Z_t\xi_t$ and using $K_{t-1}/(Z_t\xi_t) = k_{t-1}/\Gamma_t$ and $I_t/I_{t-1} = \Gamma_t i_t/i_{t-1}$,

$$\boxed{k_t = (1-\delta)\frac{k_{t-1}}{\Gamma_t} + \zeta_t\Big[1 - S\big(\tfrac{\Gamma_t i_t}{i_{t-1}}\big)\Big] i_t}, \qquad \boxed{k^u_t = \frac{\nu_t}{\Gamma_t}k_{t-1}} \tag{P-60/61}$$

where $k^u_t = K^u_t/(Z_t\xi_t) = \nu_t K_{t-1}/(Z_t\xi_t)$.

Wage inflation (59) and indexation (58). Since $W_t = w_t Z_t P_t$, gross wage inflation is $\Pi^w_t = \frac{W_t}{W_{t-1}} = \frac{w_t}{w_{t-1}}\frac{Z_t}{Z_{t-1}}\frac{P_t}{P_{t-1}} = \frac{w_t}{w_{t-1}}\Gamma^Z_t \Pi_t$, i.e. eq. (59). Consistently, the wage-indexation term (58) carries the extra $\bar{\Gamma}^Z$ (trend real-wage growth): $\Pi^{w,\text{index}}_t = \bar{\Gamma}^Z(\Pi_{t-1})^{\iota_w}(\Pi^*_t)^{1-\iota_w}$.

Non-oil marginal cost (64). In (P-33), $P^V_t/P_t = Z_t^{-(1-\alpha)}(R^k_t/\alpha)^\alpha\big(\frac{W_t}{P_t(1-\alpha)}\big)^{1-\alpha}$; substituting $\frac{W_t}{P_t} = w_t Z_t$ makes the $Z_t^{-(1-\alpha)}$ cancel against $Z_t^{1-\alpha}$, so the *real* non-oil marginal cost is stationary:

$$\boxed{p^V_t = \left(\frac{R^k_t}{\alpha}\right)^\alpha\left(\frac{w_t}{1-\alpha}\right)^{1-\alpha}} \tag{P-64}$$

Price NKPC (62). The only trend in (P-36) sits in the forward output ratio: $Y_{t+1}/Y_t = (y_{t+1}/y_t)\Gamma_{t+1}$. Hence the detrended NKPC picks up a Γ_{t+1},

$$\boxed{\frac{\Pi_t}{\Pi_t^{\text{index}}}\Big(\frac{\Pi_t}{\Pi_t^{\text{index}}}-1\Big) = \mathbb{E}_t\Big[m^s_{t,t+1}\frac{y_{t+1}}{y_t}\Gamma_{t+1}\frac{\Pi_{t+1}}{\Pi_{t+1}^{\text{index}}}\Big(\frac{\Pi_{t+1}}{\Pi_{t+1}^{\text{index}}}-1\Big)\Big] + \frac{\mu_t^p mc_t - 1}{\eta_p(\mu_t^p-1)}} \tag{P-62}$$

and the wage NKPC (56) is stationary as written (both mrs_t and w_t are Z_t-detrended).

Aggregate resource constraint (70). Dividing the levels constraint (P-70) by $Z_t\xi_t$ and using $K_{t-1}/(Z_t\xi_t) = k_{t-1}/\Gamma_t$ moves the utilization cost onto the trend:

$$\boxed{c_{N,t} + g_t + i_t + A(\nu_t)\frac{k_{t-1}}{\Gamma_t} = y_t\Big[1 - \frac{\eta_p}{2}\Big(\frac{\Pi_t}{\Pi_t^{\text{index}}}-1\Big)^2\Big]} \tag{P-70}$$

The rest of the system. Every remaining condition detrends by the same three substitutions (16): the Euler equations (45)/(50) are unchanged in form (rates and m/Π are stationary); the static consumption mixes (47)/(54), MRS (57), production block (63)–(69), oil clearing (74), GDP (75), government budget (76), fiscal/Taylor rules (77)–(79), profits (80), and the shock processes (84)–(95) all carry only stationary objects once trends are divided out. This reproduces the paper's full list (44)–(95).

I The deterministic steady state

The steady state is the object you re-target for calibration: freeze all detrended variables, set every shock to its mean ($\Gamma_t^Z = \bar\Gamma^Z$, $\Gamma_t^N = \bar\Gamma^N$ so $\Gamma_t = \bar\Gamma$; $\chi_t = \zeta_t = 1$; $\vartheta_t = \bar\vartheta$; $o_{S,t} = \bar o_S$), impose the markups at their means $\bar\mu^p, \bar\mu^w$, and set inflation at target $\Pi_t = \Pi^* = \bar\Pi$. Bars denote steady-state values. Three simplifications follow immediately:

- **Indexation collapses:** $\Pi^{\text{index}} = \bar\Pi$ and $\Pi^{w,\text{index}} = \bar\Gamma^Z\bar\Pi$, so every Rotemberg gap $(\Pi/\Pi^{\text{index}} - 1)$ and $(\Pi^w/\Pi^{w,\text{index}} - 1)$ is zero.
- **Utilization is normalized:** $\bar\nu = 1$, $A(1) = 0$, $A'(1) = \kappa_a$.
- **Adjustment costs vanish:** at $\bar\nu = 1$ and constant investment, $S = S' = 0$ (since $I_t/I_{t-1} = \bar\Gamma$, the trend around which S is centered).

Consequently the menu-cost parameters η_p, η_w, the adjustment costs ψ_i, σ_a, the indexation weights ι_p, ι_w, and all persistences/volatilities *do not enter the steady state*: they matter only for dynamics. The steady state is *block-recursive*: prices first, then quantities.

I.1 Block 1: rates, returns, and prices

From the SDFs (P-49), $\bar m^s = \bar m^w = \beta/\bar\Gamma$. The bond Euler (50) and the NKPC (P-62) then give the nominal rate and real marginal cost, and the capital block gives Tobin's Q and the rental rate:

$$\bar m^s = \frac{\beta}{\bar\Gamma}, \quad \bar R = \frac{\bar\Gamma\,\bar\Pi}{\beta\,\bar\vartheta}, \quad \bar r = \frac{\bar R}{\bar\Pi} = \frac{\bar\Gamma}{\beta\,\bar\vartheta}, \quad \overline{mc} = \frac{1}{\bar\mu^p}, \quad \bar Q^k = \frac{1}{\bar\zeta} = 1, \tag{17}$$

$$\bar R^k = \frac{\bar Q^k\big(\bar\Gamma/\beta - (1-\delta)\big)}{1-\bar\tau^k} = \frac{\bar\Gamma/\beta - (1-\delta)}{1-\bar\tau^k}, \qquad \kappa_a = (1-\bar\tau^k)\bar R^k = \bar\Gamma/\beta - (1-\delta). \tag{18}$$

The rental rate $\bar R^k$ comes from the steady-state Tobin's Q (52), $\bar Q^k = \bar m^s[(1-\bar\tau^k)\bar R^k + (1-\delta)\bar Q^k]$, solved using $\bar m^s = \beta/\bar\Gamma$; $\bar Q^k = 1$ uses the investment FOC (53) at $S = S' = 0$ with $\bar\zeta = 1$.

Note the convenience yield $\bar{\vartheta}$ drives a wedge: capital earns gross real return $\bar{\Gamma}/\beta$ while bonds earn $\bar{r} = \bar{\Gamma}/(\beta\bar{\vartheta})$.

Oil and factor prices. The oil-price observable is demeaned (obs. eq. 107), so normalize $\bar{p}^O = 1$. From the marginal-cost dual (63) with $\overline{mc} = 1/\bar{\mu}^p$,

$$\bar{p}^V = \left[\frac{\overline{mc}^{\,1-\nu_Y} - \omega_Y}{1-\omega_Y}\right]^{\frac{1}{1-\nu_Y}}, \tag{19}$$

and inverting the non-oil marginal cost (P-64) gives the real wage

$$\bar{w} = (1-\alpha)\Big(\frac{\alpha}{\bar{R}^k}\Big)^{\frac{\alpha}{1-\alpha}} \bar{p}^{V\,\frac{1}{1-\alpha}}. \tag{20}$$

I.2 Block 2: factor ratios and quantities per unit of hours

With prices fixed, all input ratios are pinned down. From the non-oil input mix (68), utilized capital (61), non-oil production (67), and the oil/non-oil mix (69) with $\bar{p}^O = 1$:

$$\frac{\bar{k}^u}{\bar{n}} = \frac{\alpha}{1-\alpha}\frac{\bar{w}}{\bar{R}^k}, \quad \bar{k} = \bar{\Gamma}\,\bar{k}^u, \quad \frac{\bar{v}}{\bar{n}} = \Big(\frac{\bar{k}^u}{\bar{n}}\Big)^{\alpha}, \quad \frac{\bar{o}_Y}{\bar{v}} = \frac{\omega_Y}{1-\omega_Y}\,\bar{p}^{V\,\nu_Y}, \tag{21}$$

and the investment–capital ratio from the law of motion (60) at $S = 0$, $\bar{\zeta} = 1$:

$$\frac{\bar{i}}{\bar{k}} = \frac{\bar{\Gamma} - (1-\delta)}{\bar{\Gamma}} \quad\Longrightarrow\quad \frac{\bar{i}}{\bar{n}} = \frac{\bar{i}}{\bar{k}}\,\bar{\Gamma}\,\frac{\bar{k}^u}{\bar{n}}. \tag{22}$$

Detrended output per hour $\bar{y}/\bar{n}$ follows from the CES production function (66). Because production is constant returns in (k^u, n) and all ratios to $\bar{n}$ are now constants, every real quantity on the production side scales linearly with hours $\bar{n}$. *Choose $\bar{n}$ as the units normalization* (equivalently, back out the disutility constant $\bar{\xi}$ in Block 4); the level of hours is not separately identified because hours enter the observables only in growth rates.

I.3 Block 3: closing quantities with the great ratios

Household oil consumption follows core consumption from the static mix (47)/(54) at $\bar{p}^O = 1$: $\bar{c}_O = \frac{\omega_C}{1-\omega_C}\,\bar{c}_N$. Two accounting identities and the great-ratio targets close the system. GDP (75) and the resource constraint (P-70) (at $\bar{\nu} = 1$, zero menu cost) are

$$\overline{GDP} = \bar{y} + \bar{p}^O\bar{c}_O = \bar{y} + \frac{\omega_C}{1-\omega_C}\bar{c}_N, \qquad \bar{c}_N + \bar{g} + \bar{i} = \bar{y}. \tag{23}$$

With $\bar{g} = s_g\overline{GDP}$, $\bar{t} = s_t\overline{GDP}$, $\bar{b}^g = s_b\overline{GDP}$ (the calibrated $\overline{G/GDP}$, $\overline{T/GDP}$, $\overline{B^g/GDP}$; the last quarterly), substitute $\bar{g}$ into the resource constraint and solve the resulting linear pair for core consumption:

$$\boxed{\bar{c}_N = \frac{(1-s_g)\,\bar{y} - \bar{i}}{1 + s_g\,\frac{\omega_C}{1-\omega_C}}} \tag{24}$$

after which $\overline{GDP}$, $\bar{g}$, $\bar{t}$, $\bar{b}^g$, $\bar{c}_O$, and (from oil clearing 74) the endowment $\bar{o}_S = \bar{c}_O + \bar{o}_Y$ are all determined.

I.4 Block 4: households, MRS, and the disutility constant

Worker bonds equal their portfolio target, $\bar{b}^w = \bar{B}^w$ (so that $\Psi' = 0$ and the worker Euler (45) coincides with the capitalist Euler (50)); $\bar{B}^w$ is set to the estimated worker bonds-to-output ratio. The worker budget (P-46) then gives worker core consumption

$$\bar{c}_N^w = \frac{1-\omega_C}{1+\tau^c}\Big[(1-\bar{\tau}^l)\frac{\bar{w}\,\bar{n}}{\lambda} + \bar{t} + \bar{b}^w \frac{1-\beta\bar{\vartheta}}{\bar{\Gamma}\bar{\Pi}}\Big], \tag{25}$$

using $\frac{1}{\bar{\Pi}\bar{\Gamma}} - \frac{1}{\bar{R}} = \frac{1-\beta\bar{\vartheta}}{\bar{\Gamma}\bar{\Pi}}$. The worker consumption bundle simplifies to $\bar{c}^w = \bar{c}_N^w/(1-\omega_C)$ at $\bar{p}^O = 1$ (the CES price index equals one), so the steady-state MRS (57) collapses to the clean form $\overline{mrs} = (1+\tau^c)\bar{\xi}(\bar{n}/\lambda)^\varphi \bar{c}^w/(1-\bar{\tau}^l)$. The steady-state wage Phillips curve (56), with the Rotemberg gaps zero, reduces to a static labor-supply condition in which the real wage carries the wage markup $\bar{\mu}^w$ (this is the exact $\eta_w \to 0$ limit of (56)/(26) derived in Section E):

$$\boxed{\bar{w} = \bar{\mu}^w(1+\varphi)\,\overline{mrs} = \bar{\mu}^w(1+\varphi)\,\frac{(1+\tau^c)\,\bar{\xi}\,(\bar{n}/\lambda)^\varphi\,\bar{c}^w}{1-\bar{\tau}^l}} \tag{26}$$

which we solve for the disutility constant $\bar{\xi}$ given the normalization of $\bar{n}$:

$$\bar{\xi} = \frac{\bar{w}\,(1-\bar{\tau}^l)}{\bar{\mu}^w(1+\varphi)(1+\tau^c)(\bar{n}/\lambda)^\varphi\,\bar{c}^w}. \tag{27}$$

Finally, capitalist consumption is the residual from aggregate consumption $\bar{c}_N = \lambda\bar{c}_N^w + (1-\lambda)\bar{c}_N^s$, i.e. $\bar{c}_N^s = (\bar{c}_N - \lambda\bar{c}_N^w)/(1-\lambda)$; capitalist bonds from bond-market clearing $\bar{b}^s = (\bar{b}^g - \lambda\bar{b}^w)/(1-\lambda)$; and profits from (80), $\bar{d} = \bar{y} - \bar{w}\bar{n} - \bar{R}^k\bar{k}^u - \bar{p}^O\bar{o}_Y$.

One equation is still unused. Given the worker budget, the resource constraint, the aggregate-consumption identity and bond-market clearing, Walras' law makes exactly *one* of the capitalist and government budgets redundant, not both. We use the remaining one, the steady-state government budget (76), to back out the consumption-tax rate:

$$\boxed{\tau^c = \frac{\bar{g} + \bar{t} + \dfrac{\bar{b}^g}{\bar{\Pi}\bar{\Gamma}} - \dfrac{\bar{b}^g}{\bar{R}} - \bar{\tau}^l\bar{w}\bar{n} - \bar{\tau}^k\big(\bar{R}^k\bar{k}^u + \bar{d}\big)}{\bar{c}_N + \bar{p}^O\bar{c}_O}} \tag{28}$$

so τ^c is an internally determined object, not a free calibration target: it is whatever rate balances the budget at the assumed spending, transfer and debt ratios. The capitalist budget then holds automatically, which is a useful numerical check. Note that τ^c, κ_a, $\bar{\xi}$ and the shock means all depend on the deep parameters, so in estimation they must be recomputed at every draw rather than fixed once.